

# $h/2e$ Flux Trapping without Superconductivity

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**Will Lancer**

*Simons Center for Geometry and Physics, Stony Brook University, Stony Brook, NY*

*E-mail:* [william.lancer@stonybrook.edu](mailto:william.lancer@stonybrook.edu)

ABSTRACT: Recently, it was shown by Komargodski and Popov [1] that ordinary metals threaded with an Aharonov–Bohm solenoid exclusively trap either 0 or  $h/2e$  flux. The latter value is familiar in the superconducting setting, yet we find that it holds in ordinary metals as well, with no condensate or pairing necessary. We provide an analytic explanation of the flux trapping, and extend the results to the case of finite temperature and time-reversal invariant potentials. We also find that the flux remains trapped at 0 and  $h/2e$  for finite temperature and for lattice regularization.

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## 1 Introduction

In the usual picture of a metal pierced by an Aharonov–Bohm solenoid [2], the magnetic flux is considered to be a fixed background for the system. In a recent paper by Komargodski and Popov [1], it was found that this view of the flux misses an interesting dynamical effect. If we allow the flux to vary, due to the back-reaction of the electrons in the system, the flux is dynamically quantized to either be 0 or  $h/2e$ . Furthermore, a similar analysis found a near-origin persistent current that discontinuously jumped as the flux was turned on.

The flux trapping is an energetic, infrared effect. It is controlled by the effective potential obtained from integrating out the fermions, which depends on the reduced flux,

$$\nu := \frac{\Phi}{\Phi_0}, \quad 0 \leq \nu < 1. \quad (1.1)$$

The effective potential  $\Omega(\nu)$  is periodic mod 1 in  $\nu$  due to large gauge invariance [3], and is even in  $\nu$  due to time-reversal symmetry. These conditions single out—in the absence of spontaneous symmetry breaking—the flux values  $\nu = 0$  and  $\nu = 1/2$ . The value chosen by the system is then energetically determined.

The value of  $\nu = 1/2$  is familiar in the superconducting setting, but here it arises for a different reason. In a superconductor, the order parameter  $\Psi$  carries charge  $2e$ , and requiring it to be single-valued quantizes flux in units of  $h/2e$ . This means that the period of the theory is itself  $h/2e$ . This is not the case here. The period of the theory remains  $h/e$ , and  $\nu = 0$  and  $\nu = 1/2$  are singled out due to their invariance under time-reversal symmetry and Byers–Yang invariance [3]. The exact details of the system’s energetic configuration then decide which value of flux is chosen.

Flux trapping fundamentally occurs because of the back-reaction of the fermions against the reduced AB flux  $\Phi$ . This back-reaction produces a current which influences the flux  $\Phi$ , which then settles into a trapped value. On a finite-disk, we can write the current as the current on an infinite-disk plus a correction term

$$j_R(r) = j_\infty(r) + \Delta j(r), \quad (1.2)$$

where as it turns out

$$\Delta j(r) = \frac{c_R(\nu)}{2\pi R} \frac{1}{r}, \quad (1.3)$$

where  $c_R$  is a constant linear in  $\nu$ . This linear correction to the current is enough to show that flux is trapped on the disk, as we demonstrate in Section 3.2.

The flux-trapping effect persists for finite temperature—it is merely damped by the Lifshitz–Kosevich [4] factor

$$D_\beta(r) = \frac{\pi^2 r / \beta}{\sinh(\pi^2 r / \beta)}. \quad (1.4)$$

Interactions do generically change the location of the flux trapping, but as long as the interactions are time-reversal invariant, the flux values of  $\nu = 0$  and  $\nu = 1/2$  are still privileged in some sense.

Flux trapping is an IR effect, and thus it should not depend on the details of the short-distance physics. To test this, we put the theory on the lattice. The lattice Hamiltonian is

$$H = \sum_i h_{ii} c_i^\dagger c_i - \sum_{ij} (e^{i\theta_{ij}} t_{ij} c_j^\dagger c_i + e^{-i\theta_{ij}} t_{ij}^* c_i^\dagger c_j), \quad (1.5)$$

where  $c_j^\dagger$  and  $c_j$  are the creation and annihilation operators,  $t_{ij}$  the amplitude for a particle to hop from site  $i$  to site  $j$ , and  $\theta_{ij}$  the Peierls phase,

$$\theta_{ij} = \frac{2\pi}{\Phi_0} \int_i^j \mathbf{ds} \cdot \mathbf{A}(\mathbf{s}). \quad (1.6)$$

The Peierls phase,  $\theta_{ij}$ , is the analog of an AB-flux on the lattice. The Peierls phase entering the Hamiltonian through the hopping terms  $t_{ij}$  is the analog of minimal coupling on the lattice.

We can then numerically simulate this model on the lattice and determine whether or not flux trapping is present. We simulate eight different values of the temperature, and

five different values of the chemical potential, as well as five different UV regularizations. This gives 200 simulations. In every simulation, either  $\nu = 0$  or  $\nu = 1/2$  flux was chosen by the system.

Flux trapping can be tested in the lab. The cylinder is the feasible target, as the disk requires phase coherence over the length scale of around 1cm. The cylinder, however, is entirely on the micron scale. Specifically, a copper cylinder with radius  $R = 1\mu\text{m}$ , height  $L = 20\mu\text{m}$ , and temperature  $T = 100\text{mK}$  is sufficient to trap flux. To allow for dynamical flux trapping, there are restrictions on the inductance of the AB solenoid. Roughly, the shorter the cylinder the softer the inductance has to be to allow for flux trapping. For the previous example, an effective inductance of 0.5nH or more is [necessary?] to trap flux.

The near-origin current is quite different from the flux trapping. It is a UV effect, and is localized at radii  $r \lesssim k_F^{-1}e^{-1/|\nu|}$ , therefore being outside the range of the effective theory of the Fermi surface. Thus, the near-origin current is non-universal, and depends closely on the precise details of the UV physics. Placing the system on the lattice destroys the persistent current. In the continuum model, the persistent current is controlled by the spectral information of the circle Dirac operator,  $D(\nu) = p_\theta - \nu$ . In Appendix C, we explore these calculations, consider the case of finite temperature, and add radial potentials to the Hamiltonian.

## 2 Trapping Flux on the Cylinder

Here we detail the general picture we will be working with. We will only deal with annular geometries in this paper, as we must pierce the metal with an Aharonov–Bohm solenoid. Specifically, we will be working with the case of a cylinder, as imaged in Figure 1. Komar-godski and Popov [1] presented numerics that show flux trapping at  $\nu = 0$  and  $\nu = 1/2$  in the case of the disk, and we do not reproduce these considerations here. Due to the piercing of the geometries by the AB-solenoid, there is naturally a gauge field present,  $A$ . We also take the metals to have possibly interacting spinless electrons living on them,  $\psi$ . These electrons back-react in response to the induced AB flux, producing the dynamic flux quantization we see in the examples below.

If our system has a gauge field,  $A$ , and the space we’re working on is not simply connected, it is natural to consider the holonomy of the gauge field around the origin,

$$\Phi = \oint A = \int_0^{2\pi} A_\theta d\theta. \quad (2.1)$$

We will find it simpler to work with the flux measured in units of the flux quantum,  $\Phi_0 = h/e$ ,

$$\nu := \frac{\Phi}{\Phi_0}, \quad 0 \leq \nu < 1. \quad (2.2)$$

Since integer holonomies can be gauged away,  $\nu$  is defined mod 1. In this paper, we will generally find it beneficial to measure things in terms of the Fermi momentum—thus, we

switch to dimensionless variables, measured as

$$x = k_F^{-1} x', \quad \hat{\psi} = k_F \hat{\psi}', \quad \mathbf{A} = k_F \frac{\hbar c}{e} \mathbf{A}', \quad \mathbf{J} = \frac{e \hbar k_F^4}{m} \mathbf{J}', \quad E = \frac{\hbar^2 k_F^2}{2m} E', \quad (2.3)$$

where  $\mathbf{J}$  is the three-dimensional electric current of the system, and  $\mathbf{A}$  the gauge field. For example, in these units,  $E'_F = 1$ , and Ampere's law is

$$\nabla' \times \mathbf{B}' = \mu'_0 \mathbf{J}', \quad (2.4)$$

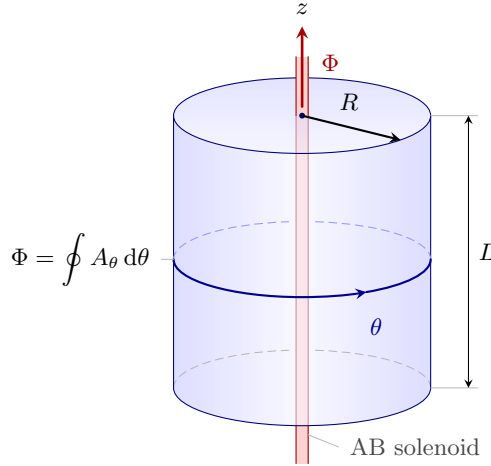
where  $\mu'_0 := 4\pi\alpha v_F/c$ . In a real metal,  $\mu'_0 \ll 1$ . This corresponds to our gauge field being very stiff, as  $\mu'_0 \ll 1$  makes the energy cost of changing the gauge field very large. In two dimensions, the dimensionless current is instead

$$\mathbf{J} = \frac{e \hbar k_F^3}{m} \mathbf{J}'. \quad (2.5)$$

The situation we're in should be clear from context. From now on, we will drop the primes on our variables.

Physically, we are working with two-dimensional objects in three-dimensional space. We encase our set-ups in a high-permeability material. This kills off any tangential  $\mathbf{B}$ -fields, thus fixing  $\hat{\mathbf{n}} \times \mathbf{B} = \mathbf{0}$  on the boundary. We will work exclusively with this set-up in our examples.

In the example below, we aim to show that—in the effective potential generated by the fermions—the value of  $\nu$  is dynamically quantized to be either 0 or 1/2. This will basically be a consequence of the energy of the system being periodic mod 1, as we will show. We will also show that the flux-trapping effect is robust to finite temperature.



**Figure 1.** The cylinder set-up. We thread a thin AB-solenoid through a cylinder of radius  $R$  and length  $L$ , with  $L \gg R$ . On the cylinder live spinless electrons,  $\psi$ . The solenoid is associated with a gauge field  $\mathbf{A}$ , and produces a flux felt by the electrons  $\Phi$ .  $\Phi$  is calculated by taking the holonomy of the gauge field,  $\Phi = \oint A = \int_0^{2\pi} A_\theta d\theta$ .

## 2.1 Trapped Flux and Time-Reversal Invariance

We now state and prove some general theorems that tell us that flux trapping at  $\nu = 0$  and  $\nu = 1/2$  is a priori expected for time-reversal invariant systems. In the case of spontaneous symmetry breaking, the ground state of the system sits at one of the values in the pair  $\{\nu_*, -\nu_*\}$  with  $\nu_*$  taken mod 1.

**Theorem 1.** *Let  $H(\nu) + V(x)$  be time-reversal invariant, with  $V(x)$  a time-reversal invariant potential, and suppose the ground-state energy  $E(\nu)$  is differentiable at  $\nu \in \{0, 1/2\}$ . Then  $\nu = 0$  and  $\nu = 1/2$  are critical points of  $E(\nu)$ .*

*Proof.* First, consider the fact that the energy—and thus the effective potential from integrating out the fermions—is periodic mod 1 in  $\nu$ . This follows from the fact that  $\nu$  itself is periodic mod 1, as integer holonomies may be gauged away. So, if  $E$  denotes the energy of our system,

$$E(\nu) = E(\nu + 1). \quad (2.6)$$

In superconductors, this fact is known as Byers–Yang invariance [3]. In ordinary metals,  $E(\nu) = E(\nu + 1)$  follows from the fact that the spectrum of the Hamiltonian remains fixed after conjugating with the large gauge transformation  $U = \exp(i \sum_j \theta_j n_j)$ , where  $\theta_j$  are phases and  $n_j$  number operators. Now, we claim that if the Hamiltonian of our system,  $H$ , is time-reversal invariant, then  $\nu = 0$  and  $\nu = 1/2$  are critical points of the energy. This follows from the fact that time-reversal is an anti-unitary symmetry, and so it acts like

$$\mathcal{T}p\mathcal{T}^{-1} = -p, \quad \mathcal{T}V(x)\mathcal{T}^{-1} = V(x). \quad (2.7)$$

In the regime of minimal-coupling, the flux enters the Hamiltonian through a term like  $(p_\theta - \nu)^2$ . Acting on such a term with  $\mathcal{T}$  gives

$$\mathcal{T}(p_\theta - \nu)^2\mathcal{T}^{-1} = (-p_\theta - \nu)^2 = (p_\theta + \nu)^2, \quad (2.8)$$

and so

$$\mathcal{T}H(\nu)\mathcal{T}^{-1} = H(-\nu). \quad (2.9)$$

Combining this with (2.6) gives the conclusion.  $\square$

Note that this conclusion only holds because  $E(\nu)$  is differentiable at the points  $\nu = 0$  and  $\nu = 1/2$ —differentiability fails in the case of the persistent current in Appendix C.

We now show that the only way for a system to sit at a point besides  $\nu \in \{0, 1/2\}$  is for  $\mathcal{T}$ -symmetry to be spontaneously broken.

**Theorem 2.** *For a time-reversal invariant Hamiltonian depending on an external flux  $\nu$ , one of the two possibilities holds:*

1. *The minima of the system lie at  $\nu = 0$  or  $\nu = 1/2$ .*

2.  $\mathcal{T}$ -symmetry is spontaneously broken, and the minimum of the system is one of the values in the pair  $\{\nu_*, -\nu_*\}$  with  $\nu_*$  taken mod 1.

*Proof.* Let  $M$  be the set of minima of the system. Since the energy is even,  $M = -M$ . Thus, if  $\nu_*$  is a minimum of  $H(\nu)$ , then  $-\nu_*$  is as well. So the minima of  $H$  appear in pairs,  $\{\nu_*, -\nu_*\}$ . These pairs fall into two categories: either  $\nu_*$  is fixed by the map  $\nu_* \xrightarrow{\mathcal{T}} -\nu_*$ , or it's not. The fixed points of this map are  $\nu_* \in \{0, 1/2\}$ , and thus these minima are invariant under time-reversal symmetry. If the point  $\nu_*$  is not fixed, then time-reversal symmetry is spontaneously broken, as the ground state of the system is not  $\mathcal{T}$ -invariant.  $\square$

So we can see that if we add interactions to the system, either the minima stay at  $\{0, 1/2\}$  or else  $\mathcal{T}$ -symmetry must be broken. Is there a way to know which minima will be chosen *prima facie*? The answer to this is unknown, but a conjecture by Leggett states that this only depends on the parity of the particle number of the system [5]. For spinless fermions in one dimension, this result has been proven by Loss [6].

## 2.2 Zero Temperature

We work on the geometry of the cylinder, as pictured in Figure 1. The system is axially symmetric, and thus  $\partial_\theta \mathbf{A}(\theta, z) = \mathbf{0}$ . To account for the back-reaction effects of the fermions, we consider the energy on the cylinder:

$$E[A_\theta(z), \psi] = \int R d\theta dz \left( \frac{1}{\mu_0 R^2} (\partial_z A_\theta(z))^2 + \left\langle p_z^2 + \frac{1}{R^2} (p_\theta - A_\theta)^2 - 1 \right\rangle \right), \quad (2.10)$$

where the  $-1$  just means that we are measuring the energy of the fermions relative to the Fermi energy  $E_F$ . Minimizing (2.10) with respect to  $\psi^*$  gives the Schrödinger equation for  $\psi$ ,

$$\left( -\partial_z^2 + \frac{(p_\theta - A_\theta)^2}{R^2} \right) \psi = E\psi. \quad (2.11)$$

Since there is axial symmetry,  $\psi(z, \theta) = e^{i\ell\theta} \psi_\ell$ . Additionally, with  $A_\theta(z) = \nu$ , there is translational symmetry in the  $z$ -direction, so for a cylinder of length  $L$ , we have that

$$\psi(z, \theta) = \frac{1}{\sqrt{2\pi RL}} e^{i\ell\theta} e^{ikz}. \quad (2.12)$$

Inputting this into (2.11) gives

$$\left( k^2 + \frac{(\ell - \nu)^2}{R^2} \right) \psi = E_{\ell,k} \psi, \quad (2.13)$$

and so the energy of the fermions is

$$E_{\ell,k} = k^2 + \frac{(\ell - \nu)^2}{R^2}. \quad (2.14)$$

We'd like to integrate out the fermions to determine the location of the trapped flux,  $\nu$ . Doing this just amounts to minimizing the grand canonical potential at fixed values of the gauge field,

$$E_{\text{eff}}[A_\theta(z) = \nu] = \min_{\psi} (E - \mu N). \quad (2.15)$$

We will work at zero temperature in this section, so  $\mu = E_F$ . In Section 2.3, we will let  $\mu$  vary. We will find that this does not change the location of the flux trapping from the set  $\nu \in \{0, 1/2\}$ . At zero temperature, integrating out  $\psi$  amounts to filling the Fermi sea. This means that we must sum over all  $\ell$  and  $k$  such that  $E_{\ell,k} < 1$ , so

$$k^2 + \frac{(\ell - \nu)^2}{R^2} \leq 1.$$

Define the cutoff momentum,

$$k_* := \sqrt{1 - \frac{(\ell - \nu)^2}{R^2}}.$$

This is real if and only if  $|\ell - \nu| \leq R$ . Filling the Fermi sea gives an energy density,  $\mathcal{E}_\psi$ ,

$$\mathcal{E}_\psi(\nu) = \frac{1}{L} E_{\text{eff}}[A_\theta(z) = \nu] = \sum_{|\ell - \nu| \leq R} \int_{-k_*}^{k_*} \frac{dk}{2\pi} (k^2 - k_*^2) = -\frac{2}{3\pi} \sum_{|\ell - \nu| \leq R} k_*^3 \quad (2.16)$$

So, we can define the effective potential

$$\Omega(\nu) := -\mathcal{E}_\psi(\nu) = \frac{2}{3\pi} \sum_{|\ell - \nu| \leq R} \left(1 - \frac{(\ell - \nu)^2}{R^2}\right)^{3/2}. \quad (2.17)$$

There are two important features of  $\Omega(\nu)$ :

1. It is symmetric under shifts  $\nu \rightarrow \nu + 1$ , as we can re-index  $\ell \rightarrow \ell - 1$ . Practically, this occurs because only the combination  $|\ell - \nu|$  appears in the summation, and thus is translationally invariant over  $\mathbb{Z}$ . This is also obvious from the fact that integer holonomies can be gauged away.
2. It is symmetric under  $\nu, \ell \rightarrow -\nu, -\ell$ . This immediately singles out two important values of  $\nu$ —0 and  $1/2$ —as these are fixed points of this operation. The function  $\Omega(\nu)$  is even around these points,  $\Omega(\nu) = \Omega(-\nu)$  and  $\Omega(1/2 + \nu) = \Omega(1/2 - \nu)$ , and thus these are also critical points of  $\Omega$ . Note that this conclusion only follows because  $\Omega$  is differentiable at these points—we will see that this is not the case for the infinite disk, and thus we cannot draw the same conclusion there.

We can understand (2.17) a bit better by summing its Fourier dual instead. Consider the function,

$$f(x) = \begin{cases} \frac{2}{3\pi} \left(1 - \frac{x^2}{R^2}\right)^{3/2} & |x| \leq R \\ 0 & \text{else} \end{cases}$$

Then, our sum (2.17) turns into

$$\Omega(\nu) = \frac{2}{3\pi} \sum_{|\ell-\nu| \leq R} \left(1 - \frac{(\ell-\nu)^2}{R^2}\right)^{3/2} = \sum_{\ell \in \mathbb{Z}} f(\ell - \nu). \quad (2.18)$$

We can then rewrite this using the Poisson summation formula, which equates the sum of a function over the integers with the sum of its Fourier transform,

$$\sum_{\ell \in \mathbb{Z}} f(\ell - \nu) = \sum_{m \in \mathbb{Z}} \tilde{f}(m) e^{-2\pi i m \nu}, \quad \tilde{f}(m) = \int_{-\infty}^{\infty} dx f(x) e^{-2\pi i m x}. \quad (2.19)$$

Rewriting (2.17) using (2.19) gives, to first order,

$$\Omega(\nu) \approx \frac{R}{4} + \frac{J_2(2\pi R)}{\pi^2 R} \cos(2\pi \nu), \quad (2.20)$$

where we used the fact that  $\tilde{f}(m)$  is real and even, and thus (2.19) is a cosine series. Each term in the  $m$ -series above goes like  $m^{-5/2}$ , and thus the first-order approximation is a good one<sup>1</sup>.

There are two important features in (2.20):

1. We may immediately conclude that  $\nu = 0$  and  $\nu = 1/2$  extremize  $\Omega(\nu)$ , depending on the sign of  $J_2(2\pi R)$ . If  $J_2(2\pi R) > 0$ , then  $\nu = 0$  is chosen—if  $J_2(2\pi R) < 0$ , then  $\nu = 1/2$  is chosen.
2. The second term in (2.20) goes like  $R^{-5/2}$  relative to the first term, and thus in the  $R \rightarrow \infty$  limit, we should expect no preferred value of flux trapping. One way to think about this is that as the radius of the cylinder becomes very large, the spacing in the angular momentum levels becomes very small,  $\Delta\ell \sim 1/R$ . Thus, the sum over  $\ell$  approximately becomes an integral over  $\ell$ , and so shifting by  $\nu$  becomes a trivial change of variables.

We can check the prediction from (2.20) numerically to confirm our results. Comparing with the numerical simulations in [1] gives perfect agreement, where  $\nu_\infty$  is the value of  $\nu$  chosen by the system:

|             | Figure 4 $\nu_\infty$ value | Predicted $\nu_\infty$ value |
|-------------|-----------------------------|------------------------------|
| $R = 1.6$   | $\nu = 0$                   | $\nu = 0$                    |
| $R = 1.7$   | $\nu = 0$                   | $\nu = 0$                    |
| $R = 1.8$   | $\nu = 0$                   | $\nu = 0$                    |
| $R = 1.9$   | $\nu = 1/2$                 | $\nu = 1/2$                  |
| $R = 2.0$   | $\nu = 1/2$                 | $\nu = 1/2$                  |
| $R = 0.125$ | $\nu = 0$                   | $\nu = 0$                    |

**Table 1.**  $\nu$  value of the maximal value of  $\Omega(\nu)$ , and the predicted  $\nu$  value of the maximal value of  $\Omega(\nu)$ . Figure 4 values taken from [1].

<sup>1</sup>One might worry that near the zeros of  $J_2(2\pi R)$ , the  $m = 2$  term dominates the  $\nu$ -dependent part of the sum, and thus introduces a new maximal  $\nu$ -value besides 0 or 1/2. This is false. Near  $R$  such that  $J_2(2\pi R) \approx 0$ ,  $J_2(4\pi R) > 0$ , and so  $\cos(4\pi \nu)$  selects for  $\nu = 0$  or  $\nu = 1/2$  yet again, *reinforcing* the trapped flux values!

### 2.3 Finite Temperature

To extend the considerations of the previous sections to finite temperature, we first must modify our computation of the grand canonical potential in (2.16). At finite- $T$ , we instead use

$$\mathcal{Z} = \text{Tr} \left( e^{-\beta(E - \mu N)} \right). \quad (2.21)$$

In the occupation number basis, this becomes

$$\mathcal{Z} = \prod_{\ell \in \mathbb{Z}} \prod_k \left( 1 + e^{-\beta(E_{\ell,k} - \mu)} \right). \quad (2.22)$$

Using

$$\Omega_{\text{gc}} = -\beta^{-1} \log(\mathcal{Z}) \quad (2.23)$$

gives

$$\Omega_{\beta}(\nu) := -\Omega_{\text{gc}} = \sum_{\ell \in \mathbb{Z}} \int_{-\infty}^{\infty} \frac{dk}{2\pi\beta} \log \left( 1 + e^{-\beta(E_{\ell,k} - \mu)} \right). \quad (2.24)$$

Define  $c_{\ell}(\nu) = (\ell - \nu)^2/R^2 - \mu$ . Performing the integral (2.24) gives a series in  $\ell$ ,

$$\Omega_{\beta}(\nu) = -\frac{1}{\sqrt{4\pi\beta^3}} \sum_{\ell \in \mathbb{Z}} \text{Li}_{3/2} \left( -e^{-\beta c_{\ell}(\nu)} \right). \quad (2.25)$$

In the  $T \rightarrow 0$  limit, we reproduce the result (2.17),

$$\lim_{\beta \rightarrow \infty} \Omega_{\beta}(\nu) \simeq \frac{2}{3\pi} \sum_{|\ell - \nu| \leq R} \left( \mu - \frac{(\ell - \nu)^2}{R^2} \right)^{3/2}, \quad (2.26)$$

with  $\mu = E_F = 1$ . We may again perform Poisson resummation on the sum (2.24). The function  $f$  we sum over is

$$f(\ell - \nu) = \int_{-\infty}^{\infty} \frac{dk}{2\pi\beta} \log \left( 1 + e^{-\beta(E_{\ell,k} - \mu)} \right). \quad (2.27)$$

This function is real and even, so computing the first two terms in the  $m$ -series gives

$$\tilde{f}(0) = -\frac{R}{2\beta^2} \text{Li}_2(-e^{\beta\mu}) \quad \tilde{f}(1) = \frac{R}{2\beta} \int_0^{\infty} dx \log \left( 1 + e^{-\beta(x - \mu)} \right) J_0(2\pi R\sqrt{x}). \quad (2.28)$$

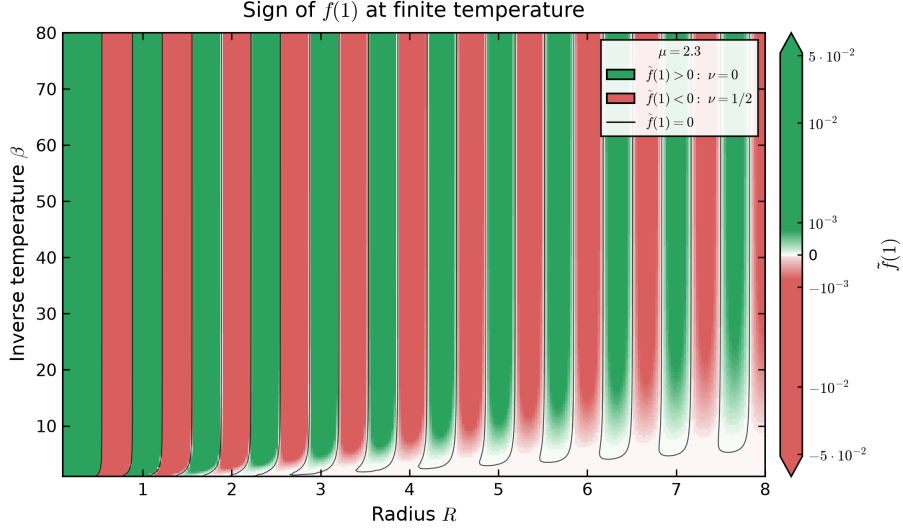
So, our effective potential becomes

$$\Omega_{\beta}(\nu) \approx \tilde{f}(0) + 2\tilde{f}(1) \cos(2\pi\nu). \quad (2.29)$$

For low temperature, we should recover (2.20). This is indeed the case, as for  $\mu = 1$  and  $\beta \gg 1$ , we have

$$\tilde{f}(0) \simeq \frac{R}{4} + \frac{\pi^2 R}{12\beta^2} \approx \frac{R}{4}, \quad (2.30)$$

$$2\tilde{f}(1) \simeq \frac{J_2(2\pi R)}{\pi^2 R} \frac{\pi^2 R/\beta}{\sinh(\pi^2 R/\beta)} \approx \frac{J_2(2\pi R)}{\pi^2 R}, \quad (2.31)$$



**Figure 2.** Heatmap of the value of  $\tilde{f}(1)$ , plotted against  $R$  and  $\beta$ . Red coloration of the graph indicates  $\tilde{f}(1) < 0$ , and thus  $\nu = 1/2$ , and green coloration indicates  $\tilde{f}(1) > 0 \iff \nu = 0$ .

which reproduces (2.20). Notice the appearance of the damping factor

$$D_\beta(R) = \frac{\pi^2 R/\beta}{\sinh(\pi^2 R/\beta)}. \quad (2.32)$$

This is the standard Lifshitz–Kosevich damping factor for oscillatory finite-temperature observables [4]. This essentially appears because most of the variation of fermionic systems occurs at the Fermi surface, with variation scale  $\beta^{-1}$ . So, if we take some zero-temperature observable  $\mathcal{O}_0$  and produce a finite-temperature observable from it,  $\mathcal{O}_T$ , this amounts to averaging the zero-temperature observable against the variation of the occupation near the Fermi surface.

We can again draw two important conclusions from (2.29):

1. The extrema of  $\Omega_\beta(\nu)$  lie at  $\nu = 0$  and  $\nu = 1/2$ , and the value that is chosen by the system is dictated by the sign of  $\tilde{f}(1)$ . If  $\tilde{f}(1) > 0$ , then  $\nu = 0$  is chosen, and if  $\tilde{f}(1) < 0$ , then  $\nu = 1/2$  is chosen. A plot of how this sign varies as we tune  $R$  and  $\beta$  is given in Figure 2, from which we can draw two conclusions. First, the value of flux trapping is essentially only determined by  $R$ , with periodicity approximately of the Bessel function zeros. Second, at large- $R$ ,  $\nu = 1/2$  is chosen much more frequently than  $\nu = 0$ . This does not matter in the  $R \rightarrow \infty$  limit, due to the relative scaling of the terms in the series.
2. Branching off of the last point, at large- $R$ , due to the relative scaling of  $\tilde{f}(1)/\tilde{f}(0) \sim R^{-5/2}$ , there is no preferred value of flux trapping as the system gets very large.

Restoring units, the Lifshitz-Kosevich factors (2.32) depend on temperature via

$$\frac{\pi^2 R}{\beta} = \frac{2\pi^2 k_B T R}{\hbar v_F}, \quad (2.33)$$

so the oscillation persists up to  $T_*(R) = \hbar v_F / 2\pi^2 k_B R$ . Note that this condition only depends on the radius of the cylinder and not its height. For a copper cylinder with  $R = 100$  nm, we get  $T = 6.1$  K, and for  $R = 1$   $\mu\text{m}$ , we get  $T = 0.61$  K [7].

There are two constraints that must be satisfied for flux trapping to occur:

1. The cylinder must be large enough to trap flux. Expanding (2.10) around a minimum  $\nu_*$  and using

$$|\Omega''(\nu_*)| = \frac{4|J_2(2\pi R)|D_\beta(R)}{R}, \quad (2.34)$$

we have

$$\ell_z = \left( \frac{4\pi}{\mu_0 R |\Omega''(\nu_*)|} \right)^{1/2} \sim \frac{\pi R^{1/4}}{\sqrt{\mu_0}}. \quad (2.35)$$

This is around  $0.11 \mu\text{m}$  for a copper cylinder with  $R = 1 \mu\text{m}$ , so for example  $L = 20 \mu\text{m}$  is sufficient to trap flux.

2. The inductance of the AB solenoid must allow for flux to dynamically trap. Precisely, changing the net flux by  $\delta\Phi$  requires driving a screening current through the solenoid, which requires an energy of  $\delta\Phi^2 / 2\mathcal{L}$ . The depth of the energy well in  $\Omega$  must be large enough to supply the energy cost of the screening current—in the worst case, the screening current requires an energy of

$$E = \frac{1}{2\mathcal{L}} \left( \frac{h}{4e} \right)^2, \quad (2.36)$$

and so we must have

$$\mathcal{L} > \frac{1}{2\Delta E} \left( \frac{h}{4e} \right)^2. \quad (2.37)$$

In practice this amounts to having  $\mathcal{L}$  be at least a few nH.

### 3 Trapping Flux on the Disk

On the finite-disk, we may take the outer radius of the disk to infinity,  $b \rightarrow \infty$ , and then ask what the value of the current density is at radius  $r$ . For a range of  $r$  values, the infinite-disk should well-approximate the finite disk of finite outer radius  $b = R$ . However, we expect at some finite radius  $r_*$  for the error term

$$\Delta j^\theta(r) = j_R^\theta(r) - j_\infty^\theta(r) \quad (3.1)$$

to become  $O(1)$  relative to the infinite current, where  $j_R^\theta(r)$  is the current density at finite outer radius  $b = R$  and  $j_\infty^\theta(r)$  is the current density with  $b \rightarrow \infty$ . In [1], there is a numerical argument that this occurs at

$$r \sim R^{1/3}. \quad (3.2)$$

We now explicitly show this to be the case, and determine that

$$r_* = \left( \frac{|2\nu - 1| \sin(\pi\nu)}{4\pi |c_R(\nu)|} \right)^{1/3} R^{1/3}, \quad (3.3)$$

where  $c_R(\nu)$  is a constant linear in  $\nu$  and depending on the fractional part of  $R/\pi$ .

Past the value of  $r_*$ , the current correction dominates, so  $j^\theta(r) = j_\infty(r) + \Delta j(r) \simeq \Delta j(r)$ . As we will show in Section 3.1, the current correction goes like  $1/r$ , and is precisely

$$\Delta j(r) = \frac{c_R(\nu)}{2\pi R} \frac{1}{r}. \quad (3.4)$$

This correction to the current is enough to show that flux is trapped on the disk, as we demonstrate in Section 3.2.

### 3.1 Infinite Disk Approximation Breakdown

First, exchange the azimuthal current for the scaled variable  $Q(r) = r^2 j^\theta(r)$ . Implement Dirichlet boundary conditions at  $b = R$ , and implement the flux as a twist,

$$\psi(\theta + 2\pi) = e^{-2\pi i \nu} \psi(\theta). \quad (3.5)$$

Then the wave functions for  $b = R$  are given by

$$\psi_{\ell,p}(r) = e^{i\ell\theta} R(r) = e^{i\ell\theta} \frac{J_{|\ell|}(j_{|\ell|,p}(r/R))}{\sqrt{\pi} R |J_{|\ell|+1}(j_{|\ell|,p})|}, \quad (3.6)$$

where  $\ell \in \mathbb{Z} - \nu$ . Computing  $j_R^\theta = (1/r^2) \text{Im}(\psi^* \partial_\theta \psi)$  from this and filling the Fermi sea gives

$$Q_R(r, \nu) = \frac{1}{\pi R^2} \sum_{\ell \in \mathbb{Z} - \nu} \ell \sum_{j_{|\ell|,p} < R} \left( \frac{J_{|\ell|}(j_{|\ell|,p}(r/R))}{|J_{|\ell|+1}(j_{|\ell|,p})|} \right)^2. \quad (3.7)$$

As computed in Appendix B,

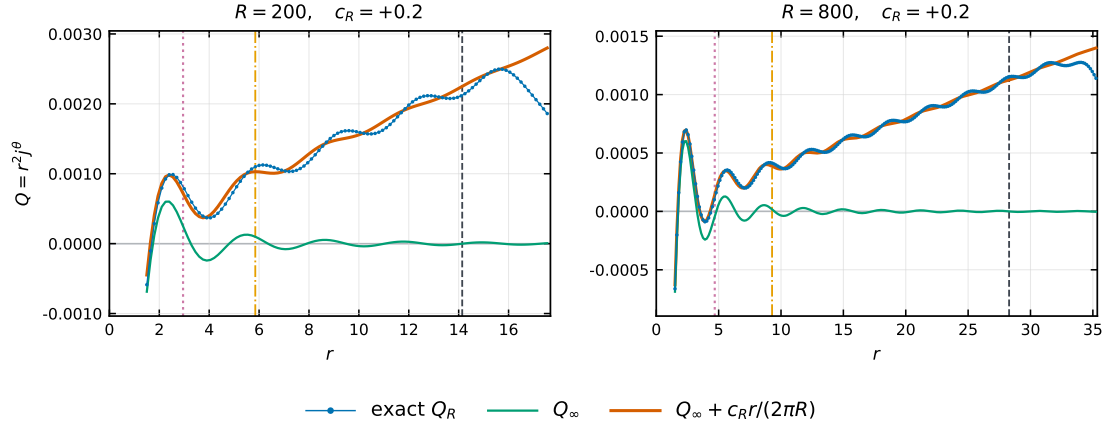
$$Q_\infty(r, \nu) = \frac{1}{2\pi} \sum_{\ell \in \mathbb{Z} - \nu} \ell \int_0^1 dk k J_{|\ell|}(kr)^2. \quad (3.8)$$

So we have the explicit error term of

$$\Delta Q(r, \nu) = Q_R(r, \nu) - Q_\infty(r, \nu). \quad (3.9)$$

The error here essentially arises from approximating the sum (3.7) by the integral (3.8). We can take the difference (3.9) by using the Euler–Maclaurin formula and by making a number of approximations. Each approximation reduces the range of validity of the formula (3.9). The details of the calculation (3.9) are in Appendix A. The result of this calculation is that

$$\Delta Q(r, \nu) = \frac{c_R}{2\pi} \frac{r}{R} + O\left(\frac{r^{2/3}}{R}\right) + o(r/R), \quad (3.10)$$



**Figure 3.** Graphs of the three different currents for two values of the outer radius. The purple dotted line is  $r_*$ , the orange dash-dotted line is  $R^{1/3}$ , and the black dashed line is  $R^{1/2}$ . The dotted blue line is the exact finite- $R$  current,  $Q_R(r)$ , the green line is the infinite- $R$  current  $Q_\infty(r)$ , and the orange line is the infinite- $R$  current plus the correction term  $\Delta Q(r) = c_R r / 2\pi R$ .

where the first error term was derived analytically, and the second only supported numerically. This approximation is valid for  $R^{1/3} \lesssim r \ll R^{1/2}$ . We plot this for multiple values of  $R$  in Figure 3.

In Appendix B, we exactly compute (3.8). It is given by

$$Q_\infty(r, \nu) = \frac{(2\nu - 1) \sin(\pi\nu) \sin(2r)}{8\pi^2 r^2} + O(r^{-3}). \quad (3.11)$$

Thus, equating (3.10) to the envelope of (3.11) gives the crossover radius,

$$r_* = \left( \frac{|2\nu - 1| \sin(\pi\nu)}{4\pi |c_R(\nu)|} \right)^{1/3} R^{1/3}. \quad (3.12)$$

### 3.2 Trapping Flux on the Disk

Consider the energy on the disk,

$$E[\nu(r)] = \int dr \left( \frac{2\pi}{\mu_0 r} (\partial_r \nu(r))^2 - \Omega(\nu(r), r) \right), \quad (3.13)$$

where  $\Omega(\nu, r)$  is the effective potential obtained from integrating out the fermions. As before,

$$\Omega(\nu, r) = -\Omega_{\text{gc}}(\nu, r), \quad (3.14)$$

where  $\Omega_{\text{gc}}$  is the grand canonical potential.  $\Omega$  obeys

$$\frac{\delta \Omega}{\delta \nu(r)} = 4\pi r j^\theta(r). \quad (3.15)$$

Varying (3.13) gives

$$\nu'' - \frac{\nu'}{r} = -\mu_0 r^2 j^\theta(r). \quad (3.16)$$

Consider  $R^{1/3} \ll r \ll R^{1/2}$ . In this regime, the current is dominated by the linear correction,

$$j^\theta(r) = j_\infty^\theta(r) + \Delta j^\theta(r) \simeq \Delta j^\theta(r), \quad (3.17)$$

where

$$\Delta j^\theta(r) = \frac{c_R(\nu(r))}{2\pi R} \frac{1}{r}, \quad (3.18)$$

as before. Write the holonomy as  $\nu(r) = \nu_* + \delta\nu(r)$ , where  $\nu_*$  is a fixed point,  $\nu_* \in \{0, 1/2\}$ . We find that  $c_R(\nu_* + \delta\nu(r)) = -\delta\nu(r)$ . Substituting this into equation (3.16) gives

$$\delta\nu'' - \frac{\delta\nu'}{r} - \frac{\mu_0}{2\pi R} r \delta\nu = 0. \quad (3.19)$$

This is solved by modified Bessel functions. Discarding a divergent solution gives the answer

$$\delta\nu(r) = C_K r K_{2/3} \left( \frac{2}{3} \left( \frac{r}{\ell_M} \right)^{3/2} \right), \quad (3.20)$$

where  $C_K$  is a coefficient and  $K_{2/3}$  a modified Bessel function of the second kind. We also introduced the length scale

$$\ell_M = \left( \frac{2\pi R}{\mu_0} \right)^{1/3}. \quad (3.21)$$

Expanding this for large  $r/\ell_M$  gives

$$\delta\nu(r) \simeq C_K \frac{\sqrt{3\pi}}{2} \ell_M \left( \frac{r}{\ell_M} \right)^{1/4} \exp \left( -\frac{2}{3} \left( \frac{r}{\ell_M} \right)^{3/2} \right). \quad (3.22)$$

The trapping length constrains the stiffness of the gauge field. We work in the regime of  $r_* \ll r \ll R^{1/2}$ , and the decay (3.20) is visible only if  $\ell_M$  obeys

$$r_* \ll \ell_M \ll R^{1/2} \iff \frac{2\pi}{R^{1/2}} \ll \mu_0 \ll \frac{2\pi}{C_*^3}, \quad (3.23)$$

where

$$r_* = C_* R^{1/3} = \left( \frac{|2\nu - 1| \sin(\pi\nu)}{4\pi |c_R(\nu)|} \right)^{1/3} R^{1/3}. \quad (3.24)$$

In a real metal,  $\mu_0 \ll 1$ , so the upper bound automatically holds. Additionally, for large- $R$  the lower bound is satisfied. The lower bound dictates the minimum size of the disk,  $R \gg (2\pi/\mu_0)^2$ . This bound is generically quite large, as  $\mu_0 = 4\pi\alpha v_F/c$  is, in real metals, around  $5 \times 10^{-4}$  [7], giving  $R \gg 10^8$  in units of  $k_F^{-1}$ . In physical units, this is a disk of a radius around a centimeter. This large size makes sense, as flux trapping is an infrared effect.

### 3.3 Lattice Regularization

We now see what happens when we replace the continuum of the infinite disk with a lattice. Due to the fact that flux trapping is an IR effect, we do not expect it to be modified by a lattice regularization. Let us recall some basic details about the lattice here. The essential ingredient behind a lattice is that we discretize space, and thus we transition from functions  $f(x)$  to finite vectors  $f(x_j) \rightarrow f_j$ <sup>2</sup>. A generic lattice Hamiltonian is then a finite matrix,  $H \rightarrow h_{ij}$ . For our purposes, we choose to work with the tight-binding model,

$$H = \sum_i h_{ii} c_i^\dagger c_i - \sum_{ij} (t_{ij} c_j^\dagger c_i + t_{ij}^* c_i^\dagger c_j), \quad (3.25)$$

where  $t_{ij}$  is the probability amplitude for a particle to hop from site  $i$  to site  $j$ . The  $c_j^\dagger$ 's create spinless fermions. Since we are working with a background AB flux, we introduce the analog of minimal coupling on the lattice. The holonomy becomes

$$W_\gamma(A) = \exp \left( \frac{2\pi i}{\Phi_0} \oint_\gamma A \right) = \exp \left( \frac{2\pi i}{\Phi_0} \int_0^{2\pi} A_\theta d\theta \right), \quad (3.26)$$

where

$$\theta_{ij} = \frac{2\pi}{\Phi_0} \int_i^j ds \cdot \mathbf{A}(s) \quad (3.27)$$

is the Peierls phase [8] associated with the hop from site  $i$  to site  $j$ . Our Hamiltonian becomes

$$H = \sum_i h_{ii} c_i^\dagger c_i - \sum_{ij} (t_{ij} e^{i\theta_{ij}} c_j^\dagger c_i + t_{ij}^* e^{-i\theta_{ij}} c_i^\dagger c_j). \quad (3.28)$$

Gauge transformations  $\zeta$  manifest as phase redefinitions on individual sites

$$c_j \mapsto e^{i\zeta_j} c_j, \quad (3.29)$$

corresponding to  $\mathbf{A} \mapsto \mathbf{A} + \nabla \zeta$ . The Hamiltonian (3.28) is gauge-invariant if  $\theta_{ij}$  transforms as

$$\theta_{ij} \mapsto \theta_{ij} + \zeta_j - \zeta_i. \quad (3.30)$$

Neither small nor large gauge transformations change the value of (3.28), as they only appear through the factors like  $e^{i\theta_{ij}}$ .

A current is the response of the Hamiltonian to a background gauge field,  $\mathbf{J} = -\delta H / \delta \mathbf{A}$ . On the lattice with our specific model (3.28), we define the current as

$$I_{ij} = \frac{2\pi i}{\Phi_0} t_{ij} (e^{i\theta_{ij}} c_j^\dagger c_i - e^{-i\theta_{ij}} c_i^\dagger c_j). \quad (3.31)$$

---

<sup>2</sup>This is for the case of a finite number of lattice sites. For an infinite, but countable, number of lattice sites, functions simply become  $f: \Lambda \rightarrow N$ , where  $\Lambda$  is the lattice and  $N$  the target space. The only essential difference then between the continuum and lattice case is the countability of the domain.

$I_{ij}$  is physically interpreted as the current going from site  $i$  to site  $j$ . For real  $t_{ij} = t$ , we have

$$\langle I_{ij} \rangle = -\frac{4\pi t}{\Phi_0} \text{Im} \left( e^{i\theta_{ij}} \langle c_j^\dagger c_i \rangle \right). \quad (3.32)$$

A lattice annulus is obtained by taking a disk, discretizing it, and then excising all lattice sites within a radius  $a$  from the origin. Similar to the continuum case, we can show that the flux values  $\nu = 0$  and  $\nu = 1/2$  are privileged in some sense. Specifically, the system will choose these values so long as time-reversal symmetry is not spontaneously broken. Via numerics, we can show that this holds for various different lattice regularizations.

**Theorem 3.** *Let  $H$  be the lattice Hamiltonian on the annulus and  $\nu$  the reduced AB flux. Then, the chosen values of  $\nu$  are 0 or  $1/2$  as long as time-reversal symmetry is not spontaneously broken.*

*Proof.* Let  $\varphi_j = \arg(x_j + iy_j) \in (-\pi, \pi]$  be the polar angle of the  $j^{\text{th}}$  site. Since  $\varphi$  has a branch cut, the raw difference  $\varphi_j - \varphi_i$  is not necessarily the angular difference between site  $i$  and site  $j$ . So, we instead define

$$\Delta_{ij} := \text{Arg} \left( e^{i(\varphi_j - \varphi_i)} \right) \in (-\pi, \pi], \quad \theta_{ij}(\nu) = \nu \Delta_{ij}. \quad (3.33)$$

Physically,  $\Delta_{ij}$  is the change in polar angle obtained from continuously following the angle along the bond. For a closed cycle  $\gamma$ ,

$$\sum_{(i,j) \in \gamma} \Delta_{ij} = 2\pi w(\gamma), \quad (3.34)$$

where  $w(\gamma) \in \mathbb{Z}$  is the winding number of  $\gamma$  around the AB solenoid. This says that contractible cycles carry zero flux, while a cycle that winds once around the puncture has holonomy  $2\pi\nu$ . Define the unitary operator

$$U = \exp \left( i \sum_j \varphi_j n_j \right), \quad (3.35)$$

where  $n_j = c_j^\dagger c_j$  is the number operator, and  $\sum_j$  sums over all lattice sites. It is simple to check that

$$U c_j U^{-1} = e^{-i\varphi_j} c_j, \quad U c_j^\dagger U^{-1} = e^{i\varphi_j} c_j^\dagger. \quad (3.36)$$

Although the real-valued angle  $\varphi_j$  jumps across its branch cut,  $e^{i\varphi_j}$  is single-valued on the lattice sites. Furthermore,

$$e^{i(\varphi_j - \varphi_i)} = e^{i\Delta_{ij}}, \quad (3.37)$$

and so conjugating a hopping term by  $U$  changes its Peierls factor from  $e^{i\nu\Delta_{ij}}$  to  $e^{i(\nu+1)\Delta_{ij}}$ . Conjugating (3.28) by (3.35) gives

$$U H(\nu) U^{-1} = H(\nu + 1), \quad (3.38)$$

and thus  $\text{spec}(H(\nu)) = \text{spec}(H(\nu + 1))$ . Applying time-reversal symmetry to (3.28) gives

$$\mathcal{T}H(\nu)\mathcal{T}^{-1} = H(-\nu), \quad (3.39)$$

and thus  $H$  is even in  $\nu$ . This follows from the fact that the only  $\nu$ -dependence in (3.28) is in the factors  $\theta_{ij}$ . By the same arguments in the proof of Theorem 2,  $\nu \in \{0, 1/2\}$  unless time-reversal symmetry is spontaneously broken.  $\square$

Since there is no magnetic field outside the AB solenoid, we can choose a vector potential that is pure gauge. Define  $\varphi$  to be the polar angle. Then, we can choose the singular gauge for  $\mathbf{A}(\mathbf{x})$ ,

$$\mathbf{A}(\mathbf{x}) = \frac{\Phi}{2\pi} \nabla \varphi, \quad \oint d\mathbf{x} \cdot \mathbf{A} = \Phi. \quad (3.40)$$

The Peierls phase becomes

$$\theta_{ij}(\nu) = \frac{2\pi}{\Phi_0} \int_i^j d\mathbf{s} \cdot \mathbf{A}(\mathbf{s}) = \nu \Delta_{ij}, \quad (3.41)$$

with  $\Delta_{ij}$  defined in (3.33). This is the phase we use in the lattice numerics below.

Now we numerically investigate flux trapping on the lattice. We take a square-lattice annulus of outer radius  $b = 16$  in units of the lattice spacing, with hoppings  $t_{ij} = t = 1$ . We take the sites to have half-integer coordinates so that the flux pierces the center of a plaquette. We use five different choices of UV data at the puncture site:

| Regularization     | Sites removed | Edge modification                  | Sites |
|--------------------|---------------|------------------------------------|-------|
| Plaquette puncture | 0             | None                               | 812   |
| Four-site hole     | 4             | None                               | 808   |
| Edge barrier       | 4             | $V = 1.5t$ on the 8 central sites  | 808   |
| Weak edge hopping  | 4             | $t' = 0.45t$ on central edge bonds | 808   |
| Larger hole        | 16            | None                               | 796   |

**Table 2.** The five different UV regularizations used in the scans below. “Central sites” refers to the eight bonds and vertices nearest the central plaquette. Adding a potential  $V$  to these sites means that we add a term  $\sum_{\text{central}} V_i c_i^\dagger c_i$  to the Hamiltonian (3.28).

For each choice, we diagonalize the Hamiltonian (3.28) on the lattice for the fundamental domain of  $0 \leq \nu \leq 1/2$ . We then compute the negative of the grand canonical potential

$$\Omega_\beta(\nu, \mu) = -\Omega_{\text{gc}} = \beta^{-1} \sum_\alpha \log \left( 1 + e^{-\beta(E_\alpha(\nu) - \mu)} \right), \quad (3.42)$$

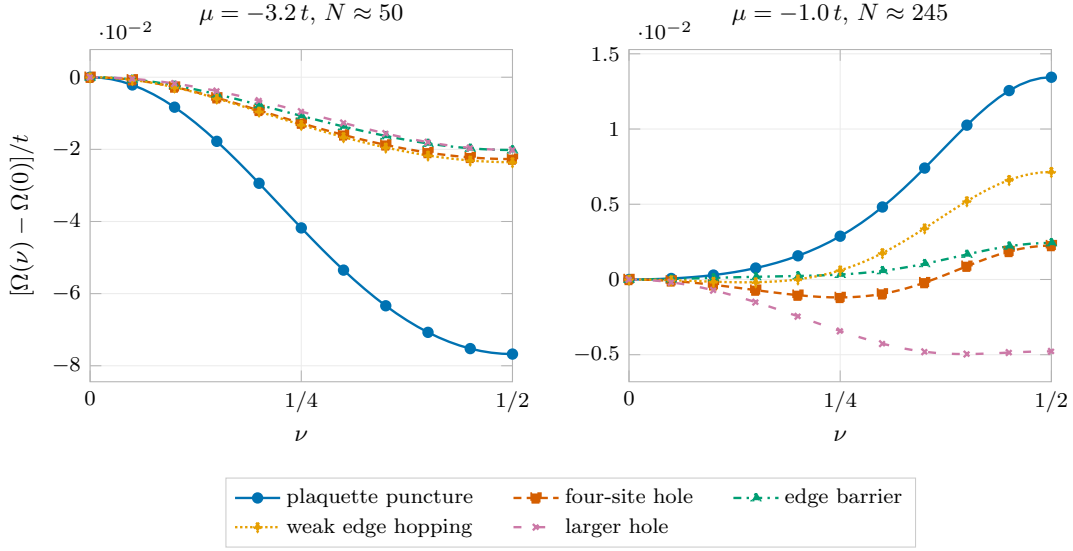
where  $E_\alpha(\nu)$  is the energy of eigenstate  $\alpha$  and  $\mu$  the chemical potential. The system will choose  $\nu$  such that  $\Omega_\beta(\nu, \mu)$  is maximized. We sample a range of  $\beta$  and  $\mu$  values—the full sweeps are listed in Table 3. As a check on the flux-periodicity of the spectrum,  $\text{spec } H(\nu + 1) = \text{spec } H(\nu)$ , the spectra of  $H(1/4)$  and  $H(5/4)$  agree to  $2 \cdot 10^{-14} t$ . Evenness

needn't be checked, as  $H(-\nu) = H(\nu)^*$ , so the equality of the spectra is exact in floating point arithmetic.

|           |                                              |
|-----------|----------------------------------------------|
| $\beta t$ | $\{1, 10, 20, 50, 70, 90, 200\}$ and $T = 0$ |
| $\mu/t$   | $\{-3.2, -2, -1, 0, 1\}$                     |

**Table 3.** Parameter sweeps used for the lattice flux-trapping scans.

Scanning over these parameters—along with the five regularizations from earlier—gives a total of 200 different simulations. These simulations confirm that the flux values  $\nu = 0$  and  $\nu = 1/2$  are trapped every time. A few representative graphs are shown below:



**Figure 4.** Representative lattice flux-trapping landscapes, plotted as  $(\Omega(\nu) - \Omega(0))$  at  $\beta t = 90$  for the five puncture regularizations. Recall that the maximum of  $(\Omega(\nu) - \Omega(0))$  dictates where the flux will be trapped. Left: at low filling,  $\mu = -3.2t$  and  $N \approx 50$ , flux is trapped at  $\nu = 0$ . Right: at higher filling,  $\mu = -1t$  and  $N \approx 245$ , the larger hole traps  $\nu = 0$  while the other regularizations trap  $\nu = 1/2$ .

## Acknowledgements

We thank F. Popov and Z. Komargodski for their support, suggestions, and for pointing out errors in earlier drafts of this paper. We also thank Claude and Codex for useful discussions. The author gratefully acknowledges financial support from the Simons Center.

## A Finite- $R$ Current Calculation

Recall that for a sufficiently smooth function  $f$  that takes values on the lattice  $\{x_n = a_0 + na\}_{n \in \mathbb{Z}}$ , with  $a$  the lattice spacing, we have

$$a \sum_{n=0}^N f(a_0 + na) = \int_{a_0}^{a_0 + Na} dx f(x) + \frac{a}{2} (f(a_0) + f(a_0 + Na)) \quad (\text{A.1})$$

$$+ \frac{a^2}{12} (f'(a_0 + Na) - f'(a_0)) + O(a^4). \quad (\text{A.2})$$

For  $r \gg k_F^{-1}$ , the Bessel zeros are approximately

$$j_{|\ell|,p} \simeq j_{|\ell|,p}^{(0)} = \pi \left( p + \frac{|\ell|}{2} - \frac{1}{4} \right) + O(\ell^2/R). \quad (\text{A.3})$$

Define  $k_{|\ell|,p} = j_{|\ell|,p}/R$ . Then  $k_{|\ell|,p+1} - k_{|\ell|,p} \simeq \pi/R$  at fixed  $|\ell|$ . So,  $a = \pi/R$  is our lattice spacing. Near the Fermi surface, where  $j_{|\ell|,p} \simeq R$ , the corrections to the Bessel zeros (A.3) are of order  $O(\ell^2/R)$ . Since only channels with  $|\ell| \lesssim r$  contribute at radius  $r$ , the range of validity of this approximation is  $r \ll R^{1/2}$ . Using this approximation requires another condition on the Bessel zeros. The radial filling condition is

$$j_{|\ell|,p} < R \iff p < \frac{1}{a} - \frac{|\ell|}{2} + \frac{1}{4}. \quad (\text{A.4})$$

This defines a maximum integer  $p$ , which we'll write as

$$P(|\ell|) = \left\lfloor \frac{1}{a} - \frac{|\ell|}{2} + \frac{1}{4} \right\rfloor, \quad (\text{A.5})$$

in the case that the argument is not an integer. If the argument is an integer, then we instead use  $P(|\ell|) = \lceil \dots \rceil - 1$ .  $P(|\ell|)$  is the upper bound on the radial sum over the Bessel zeros. The first correction to the Bessel zeros,  $j_{|\ell|,p}^{(1)} = O(\ell^2/R)$ , must obey

$$j_{|\ell|,P(|\ell|)}^{(0)} + j_{|\ell|,P(|\ell|)}^{(1)} < R, \quad j_{|\ell|,P(|\ell|)+1}^{(0)} + j_{|\ell|,P(|\ell|)+1}^{(1)} \geq R. \quad (\text{A.6})$$

This is to ensure that we are not filling the wrong number of radial levels in the sum. We will say that this is a “non-resonance” assumption, and it is actually worth discussing in a bit more detail.

The first-order correction to the Bessel zero is exactly given by

$$j_{|\ell|,p}^{(1)} = -\frac{4|\ell|^2 - 1}{8j_{|\ell|,p}^{(0)}}. \quad (\text{A.7})$$

Near the Fermi surface, the approximate and exact filling conditions disagree when the inequality (A.6) is violated, that is when

$$|R - j_{|\ell|,p}^{(0)}| \lesssim |j_{|\ell|,p}^{(1)}| = O(\ell^2/R). \quad (\text{A.8})$$

This says that for each angular-momentum channel, there is a small interval between the approximate and exact Bessel zeros within which the two filling prescriptions disagree—that is, the integer  $P(|\ell|)$  calculated from the approximate filling condition differs from the integer  $P(|\ell|)$  calculated from the exact filling condition.

Since the inequalities (A.6) are restrictions on the size of the first-order correction to the Bessel zero, they’re really conditions on the size of  $R$  in relation to the angular momentum  $|\ell|$ . So, the violation of (A.6) defines a set of “bad” outer radii,  $\mathcal{R}_{\text{bad}}(|\ell|)$ , for which the approximate and exact filling conditions disagree. Hopefully, this set of bad radii should be small in comparison to the total range of outer radii as indexed by  $|\ell|$ —this turns out to indeed be the case. We now quantify this statement.

Fix an angular momentum channel  $\alpha$ . Then we ask: what fraction of the radii at fixed  $\alpha$  are “bad”? The condition for a radius to be bad is that adding the first-order correction to the Bessel zero changes the filling level (A.6). The fraction of bad radii depends on the interval we sample from. Say we can choose radii in the interval  $(0, L]$ . Define the set of bad radii to be

$$\mathcal{R}_{\text{bad}}(\alpha) = \begin{cases} \bigcup_{p \geq 1} (j_{\alpha,p}^{(0)}, j_{\alpha,p}] & 0 \leq \alpha < 1/2 \\ \emptyset & \alpha = 1/2 \\ \bigcup_{p \geq 1} (j_{\alpha,p}, j_{\alpha,p}^{(0)}] & \alpha > 1/2. \end{cases} \quad (\text{A.9})$$

If the outer radius of our system is in this set, the filling condition changes when we add the first-order correction to the Bessel zero. The size of this set relative to the interval  $(0, L]$  is

$$\frac{|\mathcal{R}_{\text{bad}}(\alpha) \cap (0, L]|}{L} = O\left(\frac{\alpha^2 \log L}{L}\right). \quad (\text{A.10})$$

The current, however, involves a sum over all angular-momentum channels with  $|\ell| \lesssim r$ . Define the corresponding bad-radius set by

$$\mathcal{R}_{\text{bad}}^{\text{ang}}(r, \nu) = \bigcup_{\substack{\ell \in \mathbb{Z} - \nu \\ |\ell| \lesssim r}} \mathcal{R}_{\text{bad}}(|\ell|). \quad (\text{A.11})$$

The individual bad sets are strongly correlated with one another. This is because

$$j_{\alpha+2,p-1}^{(0)} = j_{\alpha,p}^{(0)}. \quad (\text{A.12})$$

So, the angular momenta split into four subfamilies:

$$\alpha = \nu + 2k, \quad 1 + \nu + 2k, \quad 1 - \nu + 2k, \quad 2 - \nu + 2k. \quad (\text{A.13})$$

Each subfamily is controlled by its largest angular momentum below  $r$ , rather than by the sum of all its channels. For fixed  $r$ ,

$$\frac{|\mathcal{R}_{\text{bad}}^{\text{ang}}(r, \nu) \cap (0, L]|}{L} = O\left(\frac{r^2 \log L}{L}\right). \quad (\text{A.14})$$

Equivalently, in a local interval near a large outer radius  $R$ , the fraction of radii for which at least one contributing channel is miscounted is

$$O\left(\frac{r^2}{R}\right). \quad (\text{A.15})$$

We can see that this fraction vanishes throughout the strict regime  $r \ll R^{1/2}$ .

At a bad radius, the incorrect filling condition can change the coefficient of the finite- $R$  correction by an order-one amount. It does not, however, change the scaling of the correction—linear scaling is retained even if a radius is bad. So, the coefficient is sensitive to bad radii, but the linear scaling of the current correction and thus the scaling of the crossover radius,  $r_*$ , remain unchanged.

For example, take  $R = 500$ ,  $\nu = 0.3$ , and  $r = R^{1/3} \simeq 7.94$ . The estimated fraction of bad radii is approximately 6.2%, while numerics give 5.9%. So, roughly 94% of nearby outer radii satisfy the non-resonance condition for every angular-momentum channel contributing at  $r$ . In the bad radius cases, the exact current differs from the predicted current by an average of 166% (this is due to switching from a small positive value to a small negative value), and flux trapping is retained in all cases.

Using the approximations above, the finite- $R$   $Q$ -term (3.7) then becomes

$$Q_R(r) \simeq \frac{a}{2\pi} \sum_{\ell \in \mathbb{Z} - \nu} \ell \sum_p k_{|\ell|,p} J_{|\ell|}(k_p r)^2, \quad (\text{A.16})$$

with  $a = \pi/R$  the lattice spacing from before. Using (A.1) gives

$$\Delta Q(r, \nu) = \frac{1}{2R} \sum_{\ell \in \mathbb{Z} - \nu} \ell \left( \frac{1}{2} - \left\{ \frac{R}{\pi} - \frac{|\ell|}{2} + \frac{1}{4} \right\} \right) J_{|\ell|}(r)^2 + \mathcal{R}_{\text{rad}}, \quad (\text{A.17})$$

where  $\{\dots\} = (\dots) - \lfloor \dots \rfloor$  denotes the fractional part of  $(\dots)$  and

$$\mathcal{R}_{\text{rad}} = \sum_{\ell \in \mathbb{Z} - \nu} \ell \mathcal{E}_{|\ell|,R} \quad (\text{A.18})$$

is the left over Euler–Maclaurin terms. Specifically,

$$\mathcal{E}_{\alpha,R} = \frac{1}{\pi R^2} \sum_{j_{\alpha,p} < R} \left( \frac{J_{\alpha}(j_{\alpha,p}(r/R))}{J_{\alpha+1}(j_{\alpha,p})} \right)^2 \quad (\text{A.19})$$

$$- \frac{1}{2\pi} \int_0^1 dk k J_{\alpha}(kr)^2 \quad (\text{A.20})$$

$$- \frac{1}{2R} \left( \frac{1}{2} - \left\{ \frac{R}{\pi} - \frac{\alpha}{2} + \frac{1}{4} \right\} \right) J_{\alpha}(r)^2. \quad (\text{A.21})$$

Define

$$\sigma_R(\alpha) = \frac{1}{2} - \left\{ \frac{R}{\pi} - \frac{\alpha}{2} + \frac{1}{4} \right\} \quad (\text{A.22})$$

to rewrite (A.17) as

$$\Delta Q(r, \nu) = \frac{1}{2R} \sum_{\ell \in \mathbb{Z} - \nu} \ell \sigma_R(|\ell|) J_{|\ell|}(r)^2 + \mathcal{R}_{\text{rad}}. \quad (\text{A.23})$$

We now make a number of approximations to be able to explicitly perform the sum (A.23).

1. Replace  $J_\alpha(r)^2$  with its Debye envelope,

$$J_\alpha(r)^2 = \frac{1}{\pi\sqrt{r^2 - \alpha^2}} + \text{oscillatory terms.} \quad (\text{A.24})$$

The oscillatory terms, when summed over  $\ell$ , give us an error term that we call  $\mathcal{R}_{\text{osc}}$ . This is not analytically bounded, but numerical evidence shows that this term scales as  $o(r/R)$ .

2. Replace the upper bound on the sum with  $M = r - r^{1/3}$  instead of  $\infty$ . The Debye approximation above is not valid when  $\alpha$  lies within  $O(r^{1/3})$  of  $r$ , as the denominator diverges when  $\alpha \rightarrow r$ . The omitted region contains  $O(r^{1/3})$  angular momentum channels, and within this region,  $|\ell| = O(r)$ . So,  $\sigma_R(|\ell|) \leq 1/2$ , and  $J_{|\ell|}(r) = O(r^{-1/3})$ . This implies that each omitted term is  $O(r^{1/3}/R)$ , and summing over the  $O(r^{1/3})$  channels gives an error term  $\mathcal{R}_{\text{turn}} = O(r^{2/3}/R)$ .
3. It is useful to pair the  $\ell = m + 1 - \nu$  and  $\ell = -m - \nu$  terms together. Doing this and using the approximations above gives the sum

$$\Delta Q(r, \nu) = \frac{1}{2R} \sum_{m < M} \left( \sigma_R(m + 1 - \nu) \frac{m + 1 - \nu}{\pi\sqrt{r^2 - (m + 1 - \nu)^2}} \right. \quad (\text{A.25})$$

$$\left. - \sigma_R(m + \nu) \frac{m + \nu}{\pi\sqrt{r^2 - (m + \nu)^2}} \right) + \mathcal{R}_{\text{osc}} + \mathcal{R}_{\text{turn}} + \mathcal{R}_{\text{rad}} \quad (\text{A.26})$$

The leading behavior of the error terms comes from  $\mathcal{R}_{\text{turn}} = O(r^{2/3}/R)$ . Define the weight function

$$w_r(\alpha) = \frac{\alpha}{\pi\sqrt{r^2 - \alpha^2}}. \quad (\text{A.27})$$

We can then approximate the weight terms in (A.25) as  $w_r(m + 1 - \nu) \approx w_r(m + \nu)$ , picking up an  $O(r^{1/3}/R)$  error term  $\mathcal{R}_{\text{shift}}$ . This gives the sum

$$\Delta Q(r, \nu) = \frac{1}{2R} \sum_{m < M} (\sigma_R(m + 1 - \nu) - \sigma_R(m + \nu)) w_r(m + \nu) + O(r^{2/3}/R) + o(r/R). \quad (\text{A.28})$$

4. We now note that the sequence

$$\zeta_R(m) = \sigma_R(m + 1 - \nu) - \sigma_R(m + \nu) \quad (\text{A.29})$$

has period 2, that is,  $\zeta(m + 2) = \zeta(m)$ . Recall that any period two sequence can be split into a constant term plus an alternating term. We write this as

$$\zeta_R(m) = c_R + (-1)^m a_R, \quad (\text{A.30})$$

with

$$c_R = \frac{\zeta(0) + \zeta(1)}{2} \quad a_R = \frac{\zeta(0) - \zeta(1)}{2}. \quad (\text{A.31})$$

Note that these constants also depend on  $\nu$ . Now, this splits the sum (A.28) into a constant and an alternating part. The alternating terms mostly cancel against each other, leaving an  $O(r^{1/3})$  error. You can see this by pairing neighboring terms in the alternating sum together and noting that since  $w_r(m)$  is increasing as a function of  $m$ , the sum of these differences is bounded by the highest term,  $O(w_r(M)) = O(r^{1/3})$ . Call the error from this term  $\mathcal{R}_{\text{alt}}$ , which scales as  $\mathcal{R}_{\text{alt}} = O(r^{1/3}/R)$ .

5. Finally, approximate the sum by an integral

$$\sum_{m < M} w_r(m) = \int_0^M dx w_r(x) + O(r^{1/3}). \quad (\text{A.32})$$

Evaluating this integral gives

$$\int_0^M dx w_r(x) = \frac{1}{\pi} (r - \sqrt{r^2 - M^2}), \quad (\text{A.33})$$

and with  $M^2 = (r - r^{1/3})^2$ , (A.33) becomes

$$\frac{1}{\pi} (r - \sqrt{r^2 - M^2}) = \frac{1}{\pi} (r - O(r^{2/3})). \quad (\text{A.34})$$

This leaves a final answer of

$$\Delta Q(r, \nu) = \frac{c_R}{2\pi} \frac{r}{R} + O\left(\frac{r^{2/3}}{R}\right) + o(r/R), \quad (\text{A.35})$$

where the scaling  $O(r^{2/3}/R)$  is analytically determined and the scaling  $o(r/R)$  is numerically supported.

## B Infinite- $R$ Current Calculation

We aim to calculate the infinite-disk current (3.8). Substituting  $u = kr$  gives

$$Q_\infty(r, \nu) = \frac{1}{2\pi r^2} \sum_{\ell \in \mathbb{Z} - \nu} \ell \int_0^r du u J_{|\ell|}^2(u). \quad (\text{B.1})$$

Define the series

$$\mathcal{J}_\nu(u) := \sum_{n=0}^{\infty} (n + \nu) J_{n+\nu}^2(u). \quad (\text{B.2})$$

Splitting the sum gives

$$Q_\infty(r, \nu) = \frac{1}{2\pi r^2} \sum_{\ell \in \mathbb{Z} - \nu} \ell \int_0^r du u J_{|\ell|}^2(u) = \frac{1}{2\pi r^2} \int_0^r du u (\mathcal{J}_{1-\nu}(u) - \mathcal{J}_\nu(u)). \quad (\text{B.3})$$

Taking the derivative of  $\mathcal{J}_\nu(u)$  and using  $2J'_\nu = J_{\nu-1} - J_{\nu+1}$  and  $(2\nu/u)J_\nu = J_{\nu-1} + J_{\nu+1}$  produces

$$\mathcal{J}'_\nu(u) = 2 \sum_{n \geq 0} (n + \nu) J_{n+\nu} J'_{n+\nu} = \frac{u}{2} \sum_{n \geq 0} (J_{n+\nu-1}^2 - J_{n+\nu+1}^2) = \frac{u}{2} (J_{\nu-1}^2 + J_\nu^2), \quad (\text{B.4})$$

where we used the fact that the series is telescoping in the last step. We may use the Lommel integral to integrate this expression, giving

$$I_\alpha(u) := \int_0^u dv v J_\alpha^2(v) = \frac{u^2}{2} (J_\alpha^2 - J_{\alpha-1} J_{\alpha+1}), \quad (\text{B.5})$$

and so

$$\mathcal{J}_\nu(u) = \frac{1}{2} (I_{\nu-1}(u) + I_\nu(u)) + C. \quad (\text{B.6})$$

The integration constant is fixed as  $C = 0$  by  $\mathcal{J}_\nu(0) = 0$ . Substituting this into our expression for  $Q_\infty(r, \nu)$  gives

$$Q_\infty(r, \nu) = \frac{1}{4\pi r^2} \int_0^r du u (I_{-\nu}(u) + I_{1-\nu}(u) - I_{\nu-1}(u) - I_\nu(u)). \quad (\text{B.7})$$

Thus our task is to evaluate expressions in the form of

$$\mathcal{I}_\alpha(r) := \int_0^r du u I_\alpha(u) = \frac{1}{2} \int_0^r du u^3 (J_\alpha^2 - J_{\alpha-1} J_{\alpha+1}). \quad (\text{B.8})$$

We can rewrite the second term using  $J_{\alpha \pm 1} = \mp J'_\alpha + (\alpha/u)J_\alpha$ . Then,

$$\mathcal{I}_\alpha(r) = \frac{1}{2} \int_0^r du u^3 \left[ J_\alpha'^2 + J_\alpha^2 \left( 1 - \frac{\alpha^2}{u^2} \right) \right]. \quad (\text{B.9})$$

We now use Bessel's equation,

$$J_\alpha'' + \frac{1}{u} J_\alpha' + \left( 1 - \frac{\alpha^2}{u^2} \right) J_\alpha = 0. \quad (\text{B.10})$$

One can check that

$$\mathcal{F}_\alpha(u) := \left( \frac{u^4}{6} + \frac{\alpha^2 u^2}{6} + \frac{\alpha^2(1 - \alpha^2)}{3} \right) J_\alpha^2 - \frac{u^3}{3} J_\alpha J'_\alpha + \left( \frac{u^4}{6} + \frac{(\alpha^2 - 1)u^2}{3} \right) J_\alpha'^2 \quad (\text{B.11})$$

satisfies

$$\frac{d}{du} \mathcal{F}_\alpha(u) = u^3 J_\alpha^2. \quad (\text{B.12})$$

For positive  $\alpha$ , the lower endpoint clearly vanishes. The dangerous cases are  $\alpha = \nu - 1$  and  $\alpha = -\nu$ , since both are in  $(-1, 0)$ . In these cases,

$$J_\alpha(u) \sim c_\alpha u^\alpha, \quad J'_\alpha(u) \sim \alpha c_\alpha u^{\alpha-1}. \quad (\text{B.13})$$

The leading terms in  $\mathcal{F}_\alpha(u)$  are then

$$\frac{\alpha^2(1-\alpha^2)}{3}c_\alpha^2 u^{2\alpha} + \frac{\alpha^2(\alpha^2-1)}{3}c_\alpha^2 u^{2\alpha} = 0, \quad (\text{B.14})$$

so the singular terms actually cancel. Thus  $\mathcal{F}_{\nu-1}(u) = O(u^{2\nu})$  and  $\mathcal{F}_{-\nu}(u) = O(u^{2-2\nu})$ , both of which vanish at the lower endpoint because  $0 < \nu < 1$ . Then,

$$\int_0^r du u^3 J_\alpha^2 = \left( \frac{r^4}{6} + \frac{\alpha^2 r^2}{6} + \frac{\alpha^2(1-\alpha^2)}{3} \right) J_\alpha^2 - \frac{r^3}{3} J_\alpha J'_\alpha \quad (\text{B.15})$$

$$+ \left( \frac{r^4}{6} + \frac{(\alpha^2-1)r^2}{3} \right) J_\alpha'^2. \quad (\text{B.16})$$

Equivalently, integrating by parts,

$$\mathcal{I}_\alpha(r) = \frac{r^2}{2} I_\alpha(r) - \frac{1}{2} \int_0^r du u^3 J_\alpha^2. \quad (\text{B.17})$$

Substituting the expression above into this identity gives the finite boundary value

$$\mathcal{I}_\alpha(r) = \frac{1}{6} \left[ (r^4 - 2\alpha^2 r^2 + \alpha^2(\alpha^2 - 1)) J_\alpha^2 + r^3 J_\alpha J'_\alpha + r^2(r^2 - \alpha^2 + 1) J_\alpha'^2 \right]. \quad (\text{B.18})$$

Here all of the Bessel functions are evaluated at  $r$ . Finally, the current may be written as

$$Q_\infty(r, \nu) = \frac{1}{4\pi r^2} (\mathcal{I}_{-\nu}(r) + \mathcal{I}_{1-\nu}(r) - \mathcal{I}_{\nu-1}(r) - \mathcal{I}_\nu(r)). \quad (\text{B.19})$$

Define  $S_\nu(r)$  as in [1]:

$$S_\nu(r) := \left( \frac{2r^2 - 2\nu^2 + 5\nu}{6} \right) J_\nu^2 + \left( \frac{2r^2 - 2\nu^2 + 3\nu - 1}{6} \right) J_{\nu+1}^2 \quad (\text{B.20})$$

$$+ \left( \frac{\nu(2\nu-1)(\nu-1)}{3r} - \frac{(2\nu+1)r}{3} \right) J_\nu J_{\nu+1}. \quad (\text{B.21})$$

We'd like to relate this to (B.19). Recalling that

$$J'_\nu = \frac{\nu}{r} J_\nu - J_{\nu+1}, \quad J_{\nu-1} = \frac{2\nu}{r} J_\nu - J_{\nu+1}, \quad (\text{B.22})$$

and using the definition of  $\mathcal{I}_\alpha(r)$  gives

$$\frac{1}{2} (\mathcal{I}_{\nu-1}(r) + \mathcal{I}_\nu(r)) = \frac{r^2}{2} S_\nu(r). \quad (\text{B.23})$$

Thus our current becomes

$$Q_\infty(r, \nu) = \frac{1}{2\pi r^2} \left[ \frac{1}{2} (\mathcal{I}_{-\nu}(r) + \mathcal{I}_{1-\nu}(r)) - \frac{1}{2} (\mathcal{I}_{\nu-1}(r) + \mathcal{I}_\nu(r)) \right] \quad (\text{B.24})$$

$$= \frac{1}{4\pi} (S_{1-\nu}(r) - S_\nu(r)). \quad (\text{B.25})$$

Since  $Q_\infty(r, \nu) = r^2 j_\infty^\theta(r)$ , raising the index with the cylindrical metric gives the current density

$$j_\infty^\theta(r) = \frac{1}{4\pi r^2} (S_{1-\nu}(r) - S_\nu(r)). \quad (\text{B.26})$$

At distances  $rk_F \gg 1$ , we can use the large- $r$  approximation for the Bessel functions. In this regime, we find that the current goes like

$$j_\infty^\theta(r) = \frac{(2\nu - 1) \sin(\pi\nu) \sin(2r)}{8\pi^2 r^4} + O(r^{-5}), \quad (\text{B.27})$$

and thus

$$Q_\infty(r, \nu) = \frac{(2\nu - 1) \sin(\pi\nu) \sin(2r)}{8\pi^2 r^2} + O(r^{-3}), \quad (\text{B.28})$$

which is (3.11).

## C UV Near-Origin Current

In this appendix, we show that there is a persistent current localized near the origin in the infinite-volume limit of the disk, at both zero and finite temperature.

Note that the above analyses necessarily miss this effect, as those were IR analyses, and this is a UV effect. The current is localized to

$$r \sim k_F^{-1} e^{-1/|\nu|}, \quad (\text{C.1})$$

which follows from the near-origin behavior of the Bessel function in (C.3). This is outside the range of the effective theory of the Fermi surface, and thus the current is non-universal.

### C.1 Zero-Temperature Disk

We now consider the case of the infinite disk, i.e. the  $a \rightarrow 0$  and  $b \rightarrow \infty$  limit of an annulus of inner radius  $a$  and outer radius  $b$ . In this limit, the wave functions become

$$\psi_{\ell, k} = \sqrt{\frac{k}{2b}} e^{i\ell\theta} J_{|\ell|}(kr), \quad (\text{C.2})$$

with  $k^2 = E$ . Calculating the current from these wave functions using  $j^\theta = (1/r^2) \text{Im}(\psi^* \partial_\theta \psi)$  gives

$$j^\theta = \sum_{\ell \in \mathbb{Z} - \nu} \sum_{E < 1} \frac{k\ell}{2br^2} (J_{|\ell|}(kr))^2. \quad (\text{C.3})$$

Computing the net current through the disk gives

$$I_{\text{tot}} = \int_0^\infty r \, dr \, j^\theta = \frac{1}{4b} \sum_{\ell \in \mathbb{Z} - \nu} \int_0^1 dE \, k \frac{dN_\ell}{dE} \text{sgn}(\ell). \quad (\text{C.4})$$

In the large-volume limit, the density of states is

$$\frac{dN_\ell}{dE} = \frac{b}{2\pi k}, \quad (\text{C.5})$$

and thus,

$$I_{\text{tot}} = \frac{1}{8\pi} \sum_{\ell \in \mathbb{Z} - \nu} \text{sgn}(\ell) = \frac{1}{8\pi} \sum_{n \in \mathbb{Z}} \text{sgn}(n - \nu). \quad (\text{C.6})$$

This sum is generically infinite and must be regularized. Before doing this, let us take a second to recognize the fact that  $I_{\text{tot}}$  is the  $\eta$ -invariant of APS. Consider the operator

$$D(\nu) = p_\theta - \nu = -i\partial_\theta - \nu \quad (\text{C.7})$$

acting on single-valued periodic functions on  $S^1$ . This one-dimensional operator is first-order and self-adjoint, and thus it's a Dirac operator. Its spectrum is

$$D(\nu)e^{in\theta} = (n - \nu)e^{in\theta}, \quad (\text{C.8})$$

and thus its eigenvalues are  $\lambda_n = n - \nu$  for  $n \in \mathbb{Z}$ . Recall the definition<sup>3</sup> of the  $\eta$ -invariant of a Dirac operator  $D$ :

$$\eta(D) = \sum_{\lambda_n} \text{sgn}(\lambda_n), \quad (\text{C.9})$$

where the sum above is regularized, as  $D$  has infinitely many eigenvalues. For  $D(\nu)$ , the  $\eta$ -invariant is given by

$$\eta(D(\nu)) =: \eta_D = \sum_{n \in \mathbb{Z}} \text{sgn}(n - \nu). \quad (\text{C.10})$$

This implies that the current may be written as

$$I_{\text{tot}} = \frac{1}{8\pi} \eta_D. \quad (\text{C.11})$$

This also gives a nice physical interpretation of the  $\eta$ -invariant: it simply counts the number of left- and right-moving angular momentum modes in the system. Regulating the sum by using an exponential-damping factor,  $e^{-\epsilon|\ell|} = e^{-\epsilon|n-\nu|}$ , gives the regulated answer

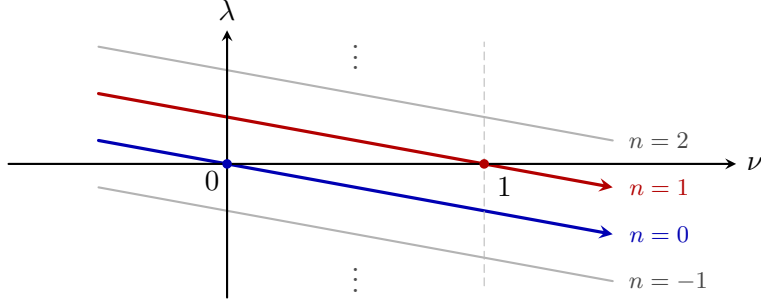
$$I_{\text{tot}} = \frac{1}{8\pi} (2\nu - 1) = \frac{1}{8\pi} \eta_D. \quad (\text{C.12})$$

An interesting fact about this current is that it discontinuously jumps from positive to negative as the flux  $\Phi$  is turned on. Precisely,

$$\Delta I_{\text{tot}} = \left( \lim_{\nu \rightarrow 0^+} I_{\text{tot}} \right) - \left( \lim_{\nu \rightarrow 0^-} I_{\text{tot}} \right) = -\frac{1}{4\pi}. \quad (\text{C.13})$$

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<sup>3</sup>We use the standard convention of  $\text{sgn}(0) = 0$ . This is also the physically sensible choice in this problem, as intuitively,  $\eta$  just counts how many left- vs. right-moving angular momentum modes we have, and  $\ell = 0$  corresponds to neither a left- nor right-moving angular momentum mode.



**Figure 5.** Spectral flow of the Dirac operator  $D(\nu)$  with eigenvalues  $\lambda_n = n - \nu$ . As  $\nu$  varies from 0 to 1, two eigenvalues touch zero: the  $n = 0$  eigenvalue goes from zero to negative, and the  $n = 1$  eigenvalue goes from positive to zero. Taking  $\nu$  mod 1, this can be seen as one eigenvalue crossing from positive to negative, thus producing a change in the  $\eta$ -invariant,  $\Delta\eta_D = -2$ .

This is actually obvious from the point-of-view of the  $\eta$ -invariant. From the definition of  $\eta_D$ , we can see that if  $\nu$  crosses an integer value, then the value of  $\eta_D$  jumps by  $\pm 2$ . Thus, crossing 0 from below produces a  $-2$  change in the value of  $\eta_D$ , which implies

$$\Delta I_{\text{tot}} = \frac{1}{8\pi} \Delta\eta_D = -\frac{1}{4\pi}. \quad (\text{C.14})$$

So, we can see that the current jump from the AB-flux crossing zero from below records the spectral flow of the Dirac operator  $D(\nu)$ , with  $\text{SF} = \Delta\eta_D/2$ . Thus, we can rewrite the above as

$$\Delta I_{\text{tot}} = \frac{\text{SF}}{4\pi}. \quad (\text{C.15})$$

A picture of  $D(\nu)$ 's eigenvalues is in figure 5.

The current value itself is situation-dependent—we will see this in the finite-temperature section below—but the jump is robust. Physically, we can interpret the current jump as a right-moving angular momentum mode crossing the Fermi energy and becoming a left-moving angular momentum mode.

Since  $I_{\text{tot}} = \partial_\nu \Omega$  (recall that  $\Omega$  is defined with a minus sign to its usual value,  $\Omega = -\Omega_{\text{gc}}$ ), we can integrate (C.12) to get

$$\Omega(\nu) = \frac{1}{8\pi}(\nu^2 - |\nu|), \quad |\nu| < 1. \quad (\text{C.16})$$

$\Omega(\nu)$  is even,  $\Omega(\nu) = \Omega(-\nu)$ , but since  $\Omega(\nu)$  has a cusp at  $\nu = 0$ , we cannot conclude that  $\nu = 0$  is a critical point. When  $\nu$  is small, (C.16) reduces to

$$\Omega(\nu) \approx -\frac{|\nu|}{8\pi},$$

and thus the current is

$$I_{\text{tot}} \approx -\frac{\text{sgn}(\nu)}{8\pi}.$$

So, we can see that there is a persistent current no matter how small the flux is! This is interesting, and may be interpreted as the system being perfectly diamagnetic at small values of  $\nu$ . This most likely only holds in the continuum model, though, and thus should be considered in that light.

We again may gain insight into this phenomenon from the perspective of  $\eta_D$ : the basic idea is that  $\eta_D$  is indifferent to the size of  $\nu$ —it only cares about  $\nu$  crossing an integer value. Since  $\Omega(\nu)$  is even, we must have opposite derivatives—and thus currents—to the left- and right-hand sides of  $\nu = 0$ . The requirement that  $\Delta\eta_D = -2$  as  $\Phi$  crosses zero from below forces persistent currents at near-zero fluxes to provide the necessary jump in the  $\eta$ -invariant. Physically, the current jump is quantized because of the discrete nature of the number of left- vs. right-moving particles in the system; turning the flux on is equivalent to turning a right-mover into a left-mover—thus producing a discrete jump in the current.

## C.2 Finite-Temperature Disk

To go from zero to finite temperature, we simply must replace the sum  $\sum_{E < 1}$  with  $\sum_E n_D(E)$ , with  $n_D(E)$  the Fermi-Dirac distribution. We have

$$I_{\text{tot}}(\beta, \mu) = \int_0^\infty r \, dr \, j^\theta = \frac{1}{4b} \sum_{\ell \in \mathbb{Z} - \nu} \int_0^\infty dE \, n_D(E) k \frac{dN_\ell}{dE} \text{sgn}(\ell). \quad (\text{C.17})$$

Performing the integral gives

$$I_{\text{tot}}(\beta, \mu) = \frac{\eta_D}{8\pi} \left( \int_0^\infty dE \, n_D(E) \right) = \frac{\eta_D}{8\pi\beta} \log(1 + e^{\beta\mu}). \quad (\text{C.18})$$

At  $\beta \rightarrow \infty$ ,  $\mu = 1$ , this reproduces the zero-temperature current (C.6). Every result of Section C.1 holds in the finite-temperature case, but they are multiplied by the factor

$$c(\beta, \mu) = \frac{\log(1 + e^{\beta\mu})}{\beta}. \quad (\text{C.19})$$

In the low-temperature limit, we have that

$$c(\beta, \mu) \approx \mu, \quad (\text{C.20})$$

and so our results are simply multiplied by  $\mu$  for large- $\beta$ . For  $\mu = 1$ ,  $\beta \rightarrow \infty$ , we clearly recover the results of Section C.1. For small- $\beta$  and fixed  $\mu$ ,

$$c(\beta, \mu) \simeq \frac{\log(2)}{\beta} + \frac{\mu}{2} + O(\beta). \quad (\text{C.21})$$

So, the current diverges as  $\beta \rightarrow 0 \iff T \rightarrow \infty$ . This is to be expected, as at fixed chemical potential  $\mu$  with  $T \rightarrow \infty$ , each energy state is half-filled, and thus the integral in (C.18) diverges. Physically, one can think of this as adding infinitely many left- and right-moving particles to the system, but in asymmetric amounts. Specifically, there is an asymmetry of  $\eta_D(\nu) = 2\nu - 1$  particles per shell, and so for an infinite number of filled shells, you get an infinite current in one direction or the other.

Integrating (C.18) with respect to  $\nu$  gives

$$\Omega(\beta, \mu) = \frac{(\nu^2 - |\nu|)}{8\pi\beta} \log(1 + e^{\beta\mu}) = \frac{c(\beta, \mu)}{8\pi} (\nu^2 - |\nu|). \quad (\text{C.22})$$

Differentiating (C.22) for small values of  $\nu$  produces

$$I_{\text{tot}}(\beta, \mu) \approx -\frac{\text{sgn}(\nu)}{8\pi} c(\beta, \mu). \quad (\text{C.23})$$

As  $\nu$  crosses zero, we see the current jump

$$\Delta I_{\text{tot}}(\beta, \mu) = \left( \lim_{\nu \rightarrow 0^+} I_{\text{tot}}(\beta, \mu) \right) - \left( \lim_{\nu \rightarrow 0^-} I_{\text{tot}}(\beta, \mu) \right) = -\frac{c(\beta, \mu)}{4\pi}. \quad (\text{C.24})$$

However, for a physical system, it's more natural to hold the particle density fixed on the disk. This produces a chemical potential that runs with  $T$ , thus fixing the divergence seen in (C.21). Precisely, we would like to fix the particle density on the disk,

$$n(\beta, \mu) = N(\beta, \mu)/\pi b^2, \quad (\text{C.25})$$

where  $N(\beta, \mu)$  is the particle number depending on  $\beta$  and  $\mu$ . Calculating  $N(\beta, \mu)$  and fixing  $n(\beta, \mu) = n_0 = 1/4\pi$  implies that  $\mu$  should run like

$$\mu(\beta) = \frac{\log(e^\beta - 1)}{\beta}. \quad (\text{C.26})$$

As  $\beta \rightarrow 0$ ,

$$\mu(\beta \rightarrow 0) \simeq \lim_{\beta \rightarrow 0} \left( -\frac{\log(1/\beta)}{\beta} + \frac{1}{2} + O(\beta) \right) = -\infty. \quad (\text{C.27})$$

This compensates for the divergence in the current, as  $c(\beta, \mu) = 1$ , and so at fixed density, eq. (C.18) becomes

$$I_{\text{tot}}(\beta, \mu) = \frac{\eta_D}{8\pi} c(\beta, \mu) = \frac{\eta_D}{8\pi}, \quad (\text{C.28})$$

regardless of the value of  $\beta$ . One way to think about this is that the current is entirely dictated by the number of particles in the system, and increasing  $T$  and decreasing  $\mu$  have antagonistic effects in this regard. Increasing  $T$  fills more energy levels with particles—thus increasing the particle number—while decreasing  $\mu$  implies that its conjugate variable,  $N$ , should also decrease to minimize the grand canonical potential. These effects work to cancel each other out to retain a constant particle density, thus keeping the current value constant.

### C.3 Radial Disorder

We now show that the persistent current is robust to adding radial potentials that are constant near the origin to the Hamiltonian. Let the Hamiltonian of our system be

$$H(r) = \frac{1}{2} \left( p_r^2 + \frac{p_\theta^2}{r^2} \right) + V(r), \quad (\text{C.29})$$

with the flux imposed through the twisted boundary condition  $\psi(r, \theta + 2\pi) = e^{-2\pi i \nu} \psi(r, \theta)$ , and with

$$V(r) = V_0 + O(r), \quad (\text{C.30})$$

as  $r \rightarrow 0$ . For example,  $V(r) = \exp(-r^2/a^2)$  or  $V(r) = (1/2)\omega^2 r^2$ . As non-examples,  $V(r) = \kappa/r^2$  and  $V(r) = \sin(1/r)$  would not satisfy the condition (C.30). The time-independent Schrödinger equation is

$$\left[ \frac{1}{2} \left( -\frac{1}{r} \partial_r (r \partial_r) - \frac{1}{r^2} \partial_\theta^2 \right) + V(r) \right] \psi(r, \theta) = E \psi(r, \theta). \quad (\text{C.31})$$

Separating variables  $\psi(r, \theta) \sim e^{i\ell\theta} R(r)$ , with  $\ell \in \mathbb{Z} - \nu$ , gives the radial equation

$$R''(r) + \frac{1}{r} R'(r) + \left( 2(E - V(r)) - \frac{\ell^2}{r^2} \right) R(r) = 0. \quad (\text{C.32})$$

Fix a self-adjoint extension at the origin by getting rid of divergent solutions, that is, those solutions to the radial equation (C.32) that diverge as  $r^{-\alpha} \rightarrow \infty$  for  $r \rightarrow 0$  with  $\alpha > 0$ . For  $\alpha = 0$ , fix the extension by discarding the logarithmic solution,  $R(r) \sim \log(r)$ . Define  $\alpha = |\ell|$ , so  $\alpha^2 = \ell^2$ . Then, we claim that the allowed solutions to (C.31) are

$$R(r) = A_{\ell, E} r^\alpha \left( 1 + \frac{V_0 - E}{2(\alpha + 1)} r^2 + O(r^3) \right), \quad (\text{C.33})$$

for  $\alpha > 0$ . This follows from the ansatz  $R(r) = r^\alpha (c_0 + c_1 r + c_2 r^2 + O(r^3))$  with  $c_0 = A_{\ell, E}$  being substituted into (C.32) with  $r$  in a neighborhood near the origin, and then matching powers of  $r$ . Since the persistent current only depends on the near-origin behavior of  $\psi(r, \theta) \sim e^{i\ell\theta} R(r)$ , adding a potential (C.30) can only change the value of the persistent current, but not the persistence of said current. Thus, the current persists at the origin, and the presence of its associated current jump is robust to the radial perturbations (C.30).

However, this really isn't a useful statement—one can simply tune  $A_{\ell, E}$  to be arbitrarily small to essentially get rid of the current. The useful statement we can extract from the above considerations is that as you turn the flux off,  $\Phi \rightarrow 0^+$ , the current localizes as a  $\delta$ -function at the origin.

To show this, let's first recall a few notions from measure theory. We say a family of measures  $d\mu_\alpha(r)$  *vaguely converges* to measure  $d\mu(r)$  if for all compactly supported test functions  $f$ , we have

$$\lim_{\alpha \rightarrow 0^+} \int_0^\infty f(r) d\mu_\alpha(r) = \int_0^\infty f(r) d\mu(r). \quad (\text{C.34})$$

We indicate vague convergence by writing  $d\mu_\alpha(r) \rightharpoonup d\mu(r)$ . The limit  $\alpha \rightarrow 0^+$  is specific to the case at hand, but we will always mean this limit when we write  $d\mu_\alpha(r) \rightharpoonup d\mu(r)$ . It is simple to show that the family of measures

$$\alpha r^{2\alpha-1} \mathbf{1}_{(0, R)}(r) dr \rightharpoonup \frac{1}{2} \delta_0, \quad (\text{C.35})$$

where  $\delta_0$  is the delta distribution measure, defined by

$$\int_0^\infty f(r) \delta_0(r) = f(0). \quad (\text{C.36})$$

Define the current measure as follows. First, define an indicator function  $\chi(r)$  that is 1 for  $r$  such that (C.33) holds and 0 otherwise. Roughly, this means that  $\chi(r) = 1$  near the origin and 0 otherwise. Then, recall that

$$j^\theta(r) = \frac{1}{r^2} \text{Im}(\psi^* \partial_\theta \psi) = \ell r^{2|\ell|-2} |A_{\ell,E}|^2 (1 + O(r^2)), \quad (\text{C.37})$$

near the origin, which follows from (C.33). Specifically, near the origin only the  $\ell = -\nu$  angular momentum mode contributes. So actually

$$j^\theta(r) = -\nu r^{2|\nu|-2} |A_{-\nu,E}|^2 (1 + O(r^2)). \quad (\text{C.38})$$

Then, define the current measure as

$$dI_{\nu,E}(r) = \chi(r) j^\theta \cdot (r dr), \quad (\text{C.39})$$

and the difference in current measures as the flux is turned off as

$$\Delta I_E(r) = \left( \lim_{\nu \rightarrow 0^+} dI_{\nu,E}(r) \right) - \left( \lim_{\nu \rightarrow 0^-} dI_{\nu,E}(r) \right). \quad (\text{C.40})$$

We claim that

$$\Delta I_E(r) = -|A_{0,E}|^2 \delta_0. \quad (\text{C.41})$$

This follows from (C.35) with  $\alpha = |\nu|$ . For  $\nu \rightarrow 0^+$ , we have

$$dI_{\nu,E}(r) \rightharpoonup -\frac{1}{2} |A_{0,E}|^2 \delta_0. \quad (\text{C.42})$$

Similarly, as the flux is turned off,  $dI_{\nu,E}(r)$  vaguely converges to  $(1/2)|A_{0,E}|^2 \delta_0$ , thus giving (C.41). So, the current localizes as a delta function near the origin as the flux is turned off. Note that the total current can still be tuned to be arbitrarily small, as it's multiplied by a factor of  $|A_{0,E}|^2$ —the  $\delta$ -function convergence is the only novel behavior that is truly robust to radial disorder.

#### C.4 Persistent Current on the Lattice

There is the question of whether or not the persistent current and its current jump appear on the lattice. As stated previously, since we're changing the UV physics, we should at the very least expect the value of the current to change, and probably its persistence to change as well.

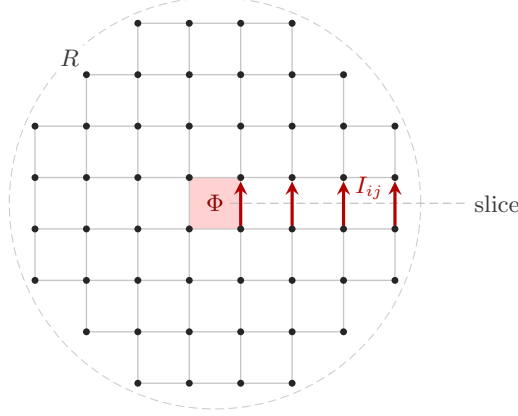
To test this, we first need to define what the analog of the persistent current is on the lattice. The analog of

$$I_{\text{tot}} = \int r dr j^\theta, \quad (\text{C.43})$$

on the lattice is defined by choosing a ray from the origin (a “slice” of the lattice), and then considering the sum of currents perpendicular to this slice,

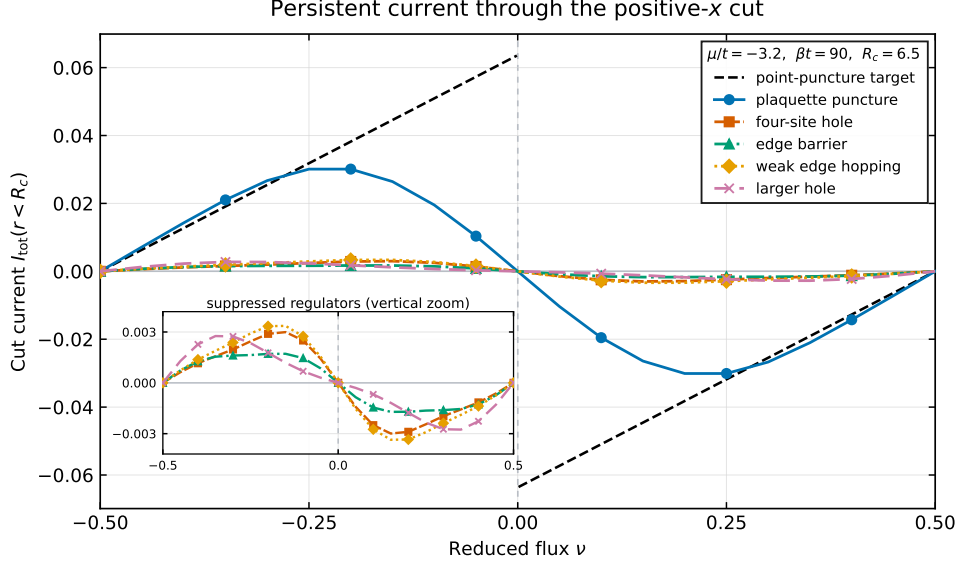
$$I_{\text{tot}} = \sum_{ij} I_{ij}. \quad (\text{C.44})$$

The sum over  $ij$  here sums over the nearest lattice sites perpendicular to the slice we’ve chosen. For example, summing over edges perpendicular to the ray  $y = 0, x > 0$  gives the lattice analog of the integral (C.43)—this construction is pictured in Figure 6.



**Figure 6.** The physical set-up on the lattice annulus, drawn for a disk of radius  $R = 4$  in lattice units. The AB solenoid threads the central plaquette with flux  $\Phi$ . The slice is the positive  $x$ -axis—the bonds crossing it carry the currents contributing to  $I_{\text{tot}}$ .

Using a slice along the positive  $x$ -axis as in Figure 6, we compute the total current  $I_{\text{tot}} = \sum_{ij} I_{ij}$ , using the same lattice regularizations and parameter sweeps above. Since the current is a short-distance phenomenon, we consider the sum up to finite radius  $R_c$ . We take  $R_c = 6.5$  in lattice units.



**Figure 7.** Total current  $I_{\text{tot}}(r < R_c)$  through the positive- $x$  slice as a function of the reduced flux  $\nu$ , plotted for the five puncture regularizations at  $\mu/t = -3.2$  and  $\beta t = 90$ , with  $R_c = 6.5$ . The current is odd in  $\nu$ , as required from  $\Omega_\beta(\nu, \mu) = \Omega_\beta(-\nu, \mu)$ . As the inner annular radius becomes wider, the current value decreases. We can see that there is no persistent current at the origin and thus no associated current jump.

From inspection of Figure 7, it's clear that the UV regularization destroys the persistent current and its associated current jump. This fact holds for 195/200 of the tested simulations. The five simulations at  $\mu = 0$  and  $T = 0$  show very small persistent currents,  $|I(\nu \rightarrow 0^+)| \sim 8 \cdot 10^{-3}$ , which is an order of magnitude below the continuum value. Thus the continuum persistent current and its current jump usually do not survive on the lattice, as was expected due to their distinction as UV effects that depend on the details of the short-range physics.

To check that this effect isn't only due to the outer boundary radius, we also tested outer radii

$$b = 12, 16, 20, 24, 28, 32, \quad (\text{C.45})$$

for  $\mu/t = -3.2$  for small values of  $\nu$ . The persistent current still vanished in these cases. There is still the question of whether or not the persistent current is present in the limit of  $b \rightarrow \infty$ .

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